# THE SPECTRE KNOT: ONE ENDLESS STRAND ON AN APERIODIC MONOTILE

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Abstract. The Spectre of Smith, Myers, Kaplan and Goodman-Strauss tiles the plane, but only non-periodically. We decorate its straight-edged form Tile(1, 1) with fourteen ribbon pieces, the same on every tile, so that ribbons continue straight across every tile edge. We prove that the pieces never close up: in every tiling by copies of Tile(1, 1) of one handedness, including tilings with faults, the ribbons contain no closed strand (the Endless Strand Theorem). In the nest tiling, grown from a single tile by repeating the substitution so that each block is child 1 (a $\Delta$-supertile) of the next, all the pieces of all the tiles join into a single strand, a two-way infinite path that passes through every tile (the Ariadne Theorem). We do not claim this for other tilings, and computations indicate that it fails in some. The proof shows that every supertile, at every level, connects the ribbon ends on its boundary in the same simple pattern: a hook at every boundary vertex but two, and one crossover. When supertiles are glued, only the parities of the shared boundary lengths matter, and these parities are the same at every level, because every side length is a Fibonacci number, twice one, or one plus or minus two. This reduces the infinite problem to finitely many cases, which we check by computer, in exact arithmetic; the geometric checks that the proof uses were done twice, by two separate programs. The proof, the programs and this text were produced with an AI system; the author has not verified every step, and the paper has not been refereed (see *Use of AI and verification status* in Section 1).

## 1. Introduction

A *Celtic knot* is a pattern of interlaced ribbons [2, 5]. How many separate strands a finite design of this kind contains has been studied, for instance, by Fisher and Mellor [7]. A *knot tile* is a polygon carrying ribbon segments that end on its edges, so that copies of the tile, placed edge to edge, link up into longer ribbons. We ask for a knot tile whose ribbons, in a tiling of the whole plane, form one endless strand. In a periodic tiling this cannot happen: if the decorated tiling is invariant under a non-zero translation, its ribbons never form a single strand (Remark 3.5). So a tile that only tiles non-periodically is the natural place to look.

In 2023 Smith, Myers, Kaplan and Goodman-Strauss described the *hat*, an aperiodic monotile [16], and then the *Spectre* [17]: a shape that tiles the plane, admits only non-periodic tilings, and admits only tilings in which all tiles have the same handedness, even when reflections are allowed [17, Thm. 2.2]. For background on tilings and substitution see [9, 1]. The Spectre is a curved-edge form of the polygon Tile(1, 1) (Fig. 1): tilings by Spectres correspond exactly to the tilings by Tile(1, 1) in which all tiles have the same handedness [17, §2].

This paper describes a knot tile on the Spectre and proves that it is *mono-stranded* in the following sense: in one explicit tiling, all its pieces form a single strand (the Ariadne Theorem), and no tiling contains a closed strand (the Endless Strand Theorem). We do not claim one strand for every tiling; computations indicate that some tilings have three (Remark 9.1). The decoration (Fig. 1) puts one crossing point on each of the fourteen edges

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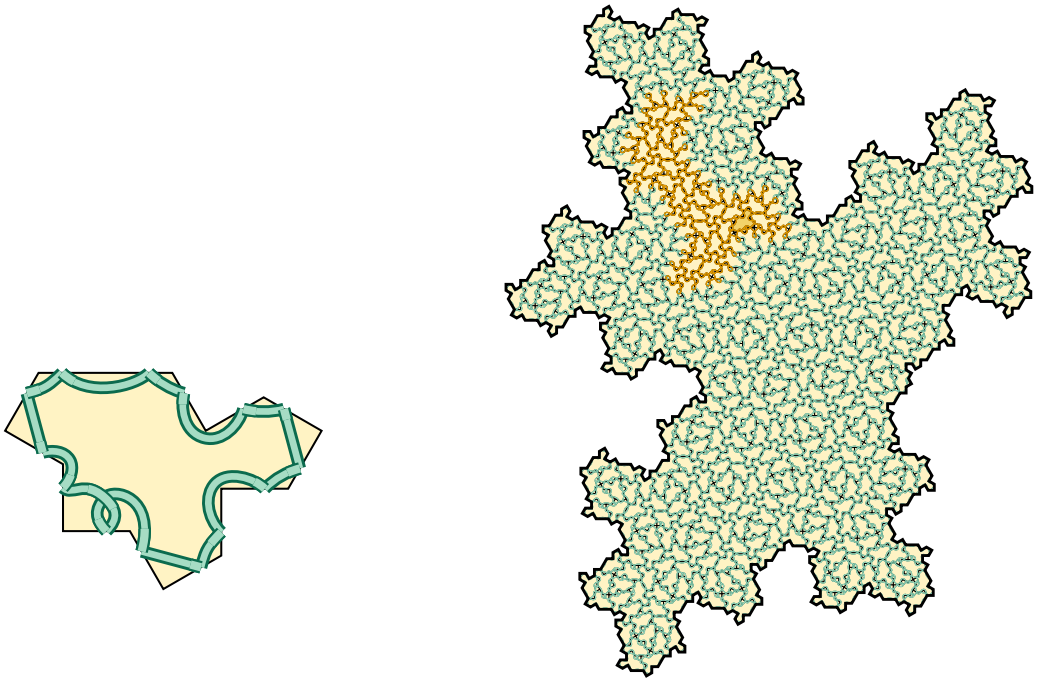


FIGURE 1. Left: the tile Tile(1, 1) with its fourteen ribbon pieces. Right: a supertile of 559 tiles (the block $N_3$ of Section 8). The original tile $N_0$ is shaded; all fourteen of its pieces already lie on a single arc of this block, drawn in orange.

and joins the 28 ribbon ends inside the tile by fourteen pieces: twelve *hooks*, each turning back round one vertex, and two pieces that cross each other inside the tile. Across an edge, the ribbons simply continue straight through the crossing point.

Our first result, the *Endless Strand Theorem*, concerns all tilings. Every tiling by Tile(1, 1) with tiles of one handedness (a *homochiral* tiling; Section 2) groups uniquely into a hierarchy of larger and larger *supertiles* [17, Thm. 2.2]. We say that the hierarchy *covers the plane* if the supertiles containing some (equivalently, every) tile exhaust the plane; equivalence holds because, once a supertile containing one tile contains another, the two tiles have the same supertiles from that level on. If it does not, the tiling splits into two or more infinite supertiles (unions of nested chains) with disjoint interiors, separated by a *fault*. (Each tile lies in exactly one supertile of each level, so two such unions are equal or disjoint.) The Endless Strand Theorem holds in both cases.

**Endless Strand Theorem.** *In every homochiral tiling of the plane by* Tile(1, 1)*, with every tile carrying the decoration of Fig. 1, there is no closed strand: every strand is a two-way infinite path of ribbon pieces.*

Our second result, the *Ariadne Theorem*, is the single strand, named after the one thread that led through the whole labyrinth. We prove it for one specific tiling, the *nest tiling* $\mathcal{N}$, which is grown from a single tile $N_0$ by repeatedly taking the $\Delta$-supertile $N_{b+1}$ that has $N_b$ as its child number 1 (Section 8). In the usual language of substitution tilings, $\mathcal{N}$ is the tiling generated from one tile as a seed by iterating the substitution about it, a fixed point of the substitution (in the combinatorial sense; compare [1]); in terms of the supertile addresses of [18], it is the infinite-level supertile whose sequence of child numbers is $1, 1, 1, \ldots$. Figures 1, 2, 9 and 10 show blocks of it.

**Ariadne Theorem.** *In the nest tiling $\mathcal{N}$, with every tile carrying the decoration of Fig. 1, all the ribbon pieces of all the tiles form a single strand. It is a two-way infinite path that passes through*

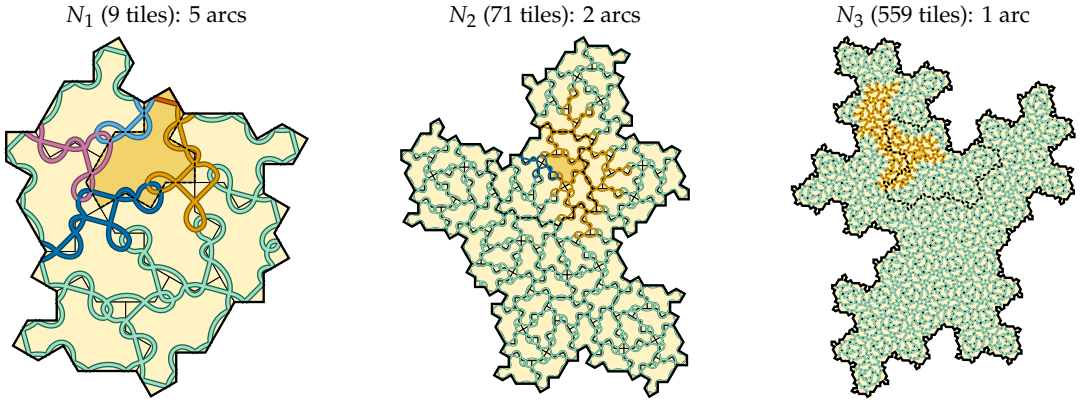



FIGURE 2. The arcs of the nest blocks $N_1$, $N_2$, $N_3$ that contain pieces of the original tile $N_0$ (shaded), one colour per arc. The dashed outline is the previous block.

*every tile. More precisely, for every $b \geq 1$, all the ribbon arcs of the block $N_b$ lie on a single arc of $N_{b+4}$, and of $N_{b+3}$ when $b$ is even.*

Figure 2 shows the merging for $N_0$: its fourteen pieces lie on 5, 2 and 1 arcs of $N_1$, $N_2$ and $N_3$. Since the curves cross only transversally and in pairs, over and under can be assigned alternately, as in Celtic knotwork (Remark 3.6); the single strand then becomes an alternating diagram of a line.

**The method.** The proof has three ideas.

(1) *Black boxes.* Seen from outside, a supertile is just a list of which ribbon end on its boundary is joined to which. We show that every supertile of every kind and level has the same *signature*: a hook at every boundary vertex but two adjacent ones, $A$ and $B$, where the two ribbons cross over; and no closed strand inside (Definition 4.3, Theorem 7.3).
(2) *Zigzags and parity.* When two such boxes are glued along a common boundary path, the ribbons zigzag along it, and the outcome depends only on the parity of its length (Lemma 5.1). Gluing therefore reduces to a finite walk on a small combinatorial object, the *skeleton* (Section 5).
(3) *Fixed parities.* The supertiles are combinatorially hexagons and compose by the substitution of [17, Fig. 5.1]. Their side lengths are given by Fibonacci numbers (Proposition 6.2), and their parities are the same at every level. So, for each kind, the skeleton takes only two values, one for odd and one for even levels, and an induction runs for ever.

The merging of all ribbons into one strand (Theorem 8.5) follows the same pattern. The label bookkeeping depends on the level only through finitely many comparisons of side lengths. We certify these for all levels at once, using the Fibonacci form.

**Computer assistance.** The proof uses computer checks of finitely many explicit facts: the signature of the Mystic and the level-1 data (vertex positions, side lengths, the position of $A$), the base cases (levels 1 to 3), the agreement of our supertile data with [17, Figs. 4.1, 5.1], the merging for the first nest block, and a symbolic run of the merging bookkeeping that produces 1,407 distinct linear conditions on side lengths, each of which is certified. The geometric computations that the proof uses (the Mystic and level-1 data, the base cases, the composition pattern for $m \leq 3$, and the merging for $b \leq 3$) were carried out twice, by two separate programs: once in JavaScript with floating-point coordinates, and once in Python in exact arithmetic, with coordinates in $\mathbb{Z}[e^{i\pi/6}]$. The skeleton walks, the symbolic run and the certificates are exact computations with integers and rationals. The reduced walk has one implementation. The symbolic run has two: the second was written separately, in Python,

from the definitions of this paper. Both agree with the direct computations wherever both apply. We also checked the conclusions directly on the actual supertiles up to level 9 (about $1.3 \times 10^8$ tiles). Appendix A lists every computation and the script that performs it. For tilings with faults, the Endless Strand Theorem also uses a finite computation on the ways in which two or three supertiles can meet. The programs, with logs of a full run, are published with this paper at `https://cortado-j.github.io/SPECTRE_KNOT/` (folder `code/`).

**Scope.** The Endless Strand Theorem covers every homochiral tiling, including tilings with faults. The Ariadne Theorem is proved for the nest tiling only, and we make no claim about the number of strands in any other tiling. The single strand is not a property of every tiling: computations indicate that in at least one other tiling, whose hierarchy covers the plane, the pieces form exactly three strands (Remark 9.1). See Section 9.

**Related work.** Curves drawn on tiles and joined across edges appear in several settings; in all of them each curve is closed or two-way infinite, as in Lemma 3.4. For Truchet tilings by squares carrying two quarter-circle arcs, the closed curves of random tilings have been studied by percolation and renormalisation methods [14, 10]. On rectangular grids, the mirror curves behind Celtic, Tchokwe and Tamil designs have a computable number of components; without internal mirrors it is the greatest common divisor of the side lengths [11]. On aperiodic tilings, the arc decoration of the Penrose rhombs due to Conway and Penrose gives closed curves apart from at most two that cross the whole tiling (see [8] and the references there), and Girih interlace drawn on Penrose tilings appears to consist of closed strands of unbounded variety [4]. Single curves through all the tiles of finite patches have been built as Hamiltonian cycles on Ammann–Beenker tilings [15], and as space-filling self-similar curves from decorations of aperiodic substitution tilings [13]. Closest to the present work, an online project [3] decorates selected edges of Tile$(1, 1)$ and describes, in one nested tiling, a single non-crossing curve that visits every tile once, with a machine check of its combinatorial core. That curve uses only some of the edges and has no crossings. Here every edge of every tile carries a ribbon end, the pieces cross, and the single strand is a knot diagram in the sense of Celtic knotwork. We are not aware of earlier work on such full knotwork decorations that form a single strand.

**Use of AI and verification status.** The author came up with the original idea of looking for a mono-stranded aperiodic knot, and then used an AI system (Claude, by Anthropic) to search for it. The search produced 34 candidate designs; the author chose design 0.1:01 among them and drew its ribbons by hand in a tile editor. At the author's direction the AI system also wrote all the programs, developed the proof and drafted this paper.

The author has not checked every step of the proof, and parts of it are beyond the author's expertise. The paper has not been refereed, and no mathematician has reviewed it, so its main result should be regarded as a claim, not as an established theorem. The AI system reviewed the argument in several passes, acting as a referee, and the errors it found were corrected. All the computations have been re-run in full; the programs and their logs are published with the paper. Where this paper calls a program or a check *independent* or *separate*, it means a separate implementation; all the programs were written by the same AI system.

**Outline.** Section 2 recalls the facts about Spectre tilings that we use, in the notation of [17]. Section 3 defines the decoration and its strands. Section 4 introduces boxes and the signature. Section 5 proves the zigzag lemma and the reduction to skeletons. Section 6 computes the side lengths. Section 7 proves the signature at every level and the Endless Strand Theorem. Section 8 constructs the nest tiling and proves the Ariadne Theorem. Section 9 discusses scope and open questions.

## 2. Spectre tilings

2.1. **The tile.** Let $\rho = e^{i\pi/6}$. The polygon Tile(1, 1) has fourteen edges of unit length; we label its vertices $v_0, \dots, v_{13}$ counter-clockwise and its edges $e_i = v_i v_{i+1}$ (indices mod 14), with $v_0 = 0$ and

$$v_{i+1} - v_i = \rho^{k_i}, \qquad (k_0, \dots, k_{13}) = (0, 10, 1, 3, 0, 2, 5, 7, 4, 6, 6, 8, 11, 9).$$

All vertices lie in $\mathbb{Z}[\rho]$, and the edge directions are multiples of 30°; see Fig. 5. This is the equilateral member Tile(1, 1) of the family Tile($a, b$) [17, §§1–2]; as there, its long side is split into the two collinear edges $e_9, e_{10}$, so that $v_{10}$ has interior angle 180°. Our Tile(1, 1) is the mirror image of the polygon drawn in [17, Fig. 1.1]; it has the handedness of the single Spectre and of the Mystic drawn in [17, Fig. 2.1]. *Spectres* are tiles, obtained for instance by replacing every edge of Tile(1, 1) by a suitable curve, whose tilings correspond exactly to the homochiral tilings by Tile(1, 1) [17, §2, Lemma 2.1]. We work throughout with Tile(1, 1) and *homochiral* tilings, in which any two tiles are related by a rotation and a translation; reflecting the whole plane (with the decoration below, which maps strands to strands) if necessary, we assume that every tile is the image of Tile(1, 1) under a rotation and a translation. Since the correspondence keeps the isometries $g_i$ [17, §2], and [17, p. 5] treats the two as the same universe of tilings, what [17] proves for tilings by Spectres holds for homochiral tilings by Tile(1, 1), and we use it in that form. Such tilings exist [17, Thm. 2.2], and every tiling by Tile(1, 1) is vertex to vertex, so tiles meet full edge to full edge [17, proof of Lemma 2.1]. The tile has no non-trivial symmetry, proper or improper (its cyclic sequence of edge directions has none), so an isometry between two tiles is unique.

2.2. **Supertiles.** We use the hierarchy of [17]. There are nine *kinds*

$$\mathcal{K} = \{\Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi\}.$$

A *level*-0 *supertile* of kind $K \neq \Gamma$ is a single tile; a level-0 supertile of kind $\Gamma$ is a *Mystic* [17, Fig. 2.1], the pair of tiles $\{\text{Tile}(1,1),\ g(\text{Tile}(1,1))\}$ with $g(z) = \rho z + v_8$, and its congruent copies; we call Tile(1, 1) its *first* tile. (The outline of the Mystic, which [17] calls symmetric, has a mirror symmetry, but the Mystic has no non-trivial proper symmetry: a proper symmetry exchanging its tiles would be an involution taking Tile(1, 1) to $g(\text{Tile}(1,1))$, but the only proper motion doing so is $g$, a rotation by 30°. So the first tile is determined by the pair.) A level-$(m+1)$ supertile is the union of level-$m$ supertiles, its *children*, arranged in a fixed way: eight children, or seven for kind $\Gamma$, of the kinds listed in Table 1. A level-1 supertile has the kind of its cluster in [17, Fig. 4.1], which is determined by the positions of its six vertices (Section 2.3) [17, proof of Thm. 4.1]; a level-$(m+1)$ supertile, $m \geq 1$, has kind $T$ if its children form the panel $T$ of [17, Fig. 5.1]. For $m \geq 1$, child $c$ is the child at the position in the last row of Table 1; at level 1 the children are numbered as in Fig. 3. (This is the order used by Kaplan's implementation of the key-point construction of [17, App. A], see [12].) For definiteness a single tile gets the kind of its position in its level-1 supertile; only the distinction between $\Gamma$ and the other kinds is used at level 0. In particular child 1 of every supertile is a $\Delta$, and the last child is a $\Gamma$; at levels $m \geq 1$ a $\Delta$ occurs only as child 1.

The nine level-1 supertiles are the Spectre clusters (9 tiles) and the Mystic cluster (8 tiles, kind $\Gamma$) of [17, Fig. 2.1], with the markings of [17, Fig. 4.1], up to reflection: [17, Fig. 2.1] draws the clusters with tiles of the opposite handedness to its single Spectre, and so to Tile(1, 1). They are drawn, with tiles of the handedness of Tile(1, 1), in Fig. 3. We number levels so that our level-$m$ supertiles are made of level-$(m-1)$ supertiles. In the language of [17], the level-1 supertiles are the clusters, which are combinatorially the marked hexagons of [17, Fig. 4.2], and the level-$n$ supertiles of hexagons of [17, §5] are our level-$(n+1)$ supertiles.

The following theorem restates, in our language, results of [17].

**Theorem 2.1** (Smith, Myers, Kaplan, Goodman-Strauss [17])**.** *Let $\mathcal{T}$ be a homochiral tiling by* Tile(1, 1).

| parent $T$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $\Gamma$ | $\Pi$ | $\Delta$ | $\Theta$ | $\Sigma$ | $\Xi$ | $\Phi$ | $\Gamma$ | |
| $\Delta$ | $\Xi$ | $\Delta$ | $\Xi$ | $\Phi$ | $\Sigma$ | $\Pi$ | $\Phi$ | $\Gamma$ |
| $\Theta$ | $\Psi$ | $\Delta$ | $\Pi$ | $\Phi$ | $\Sigma$ | $\Pi$ | $\Phi$ | $\Gamma$ |
| $\Lambda$ | $\Psi$ | $\Delta$ | $\Xi$ | $\Phi$ | $\Sigma$ | $\Pi$ | $\Phi$ | $\Gamma$ |
| $\Xi$ | $\Psi$ | $\Delta$ | $\Pi$ | $\Phi$ | $\Sigma$ | $\Psi$ | $\Phi$ | $\Gamma$ |
| $\Pi$ | $\Psi$ | $\Delta$ | $\Xi$ | $\Phi$ | $\Sigma$ | $\Psi$ | $\Phi$ | $\Gamma$ |
| $\Sigma$ | $\Xi$ | $\Delta$ | $\Xi$ | $\Phi$ | $\Sigma$ | $\Pi$ | $\Lambda$ | $\Gamma$ |
| $\Phi$ | $\Psi$ | $\Delta$ | $\Psi$ | $\Phi$ | $\Sigma$ | $\Pi$ | $\Phi$ | $\Gamma$ |
| $\Psi$ | $\Psi$ | $\Delta$ | $\Psi$ | $\Phi$ | $\Sigma$ | $\Psi$ | $\Phi$ | $\Gamma$ |
| position in [17, Fig. 5.1] | L | D | B | R | S | TR | TL | G |

TABLE 1. The children of each kind of supertile. For $T = \Gamma$ there is no child at position B, and the positions of children $2, \dots, 6$ are R, S, TR, TL, G. The positions name the places in the eight-hexagon layout of [17, Fig. 5.1]: L left, D the $\Delta$, B bottom, R right, S the $\Sigma$, TR top right, TL top left, G the $\Gamma$.

(a) [17, Thms. 2.2, 3.1, 4.1, §5, Fig. 2.1] *The tiles of $\mathcal{T}$ group uniquely into level-1 supertiles. Each contains one Mystic, in the position shown in* [17, Fig. 2.1]*; this Mystic and the other tiles, taken singly, are the level-0 supertiles inside it. For every $m \geq 1$ the level-$m$ supertiles of $\mathcal{T}$ group uniquely into level-$(m+1)$ supertiles. Each tile lies in a unique nested sequence $X_0 \subset X_1 \subset \cdots$ with $X_m$ a level-$m$ supertile.*

(b) [17, Thms. 3.1, 4.1, 5.1, §5] *For every $m \geq 1$, the tiling of the plane by the level-$m$ supertiles of $\mathcal{T}$ is combinatorially equivalent (homeomorphic as a topological complex* [17, §2.2]*) to a tiling by regular hexagons, each level-$m$ supertile of kind $K$ corresponding to a hexagon marked as the hexagon $K$ of* [17, Fig. 4.2] *for even $m$, and as its mirror image for odd $m$, adjacent supertiles corresponding to hexagons sharing an edge with matching labels; and the grouping into level-$(m+1)$ supertiles corresponds, up to reflection, to the grouping of hexagons shown in* [17, Fig. 5.1].

In (a), [17, Thm. 4.1] gives the grouping into clusters; its uniqueness, like that of the higher levels, is part of [17, Thm. 2.2], and we use it only to speak of *the* hierarchy of $\mathcal{T}$. In (b), the handedness is that of our level-1 supertiles, which are the mirror images of the clusters drawn in [17, Figs. 2.1, 4.1] (see the base case of $(*)$ in the proof of Proposition 2.4); it alternates with $m$ by [17, Thm. 5.1].

In particular every level-$m$ supertile ($m \geq 1$) is a closed topological disc whose boundary, its *rim*, is a simple closed polygon made of tile edges. Its *vertices* are the six points of the rim where it meets two other level-$m$ supertiles, and its *sides* are the six arcs of the rim between consecutive vertices. (The tile vertices on a rim are called *rim vertices*; for a supertile of level $\geq 1$, "vertex" alone always means one of the six.) A vertex of one supertile that lies on the rim of another is a vertex of that one too, since exactly three supertiles meet there. Each side is shared, as a path of tile edges, with exactly one neighbouring supertile.

2.3. **Own orientation, side numbering and the crossover corner.** By [17, Thm. 5.1], the supertiles composed from a tiling by marked hexagons are combinatorially equivalent to the *mirror images* of those hexagons. So, at each level, the cyclic order of the labelled sides is reversed, and it is convenient to orient supertiles alternately.

**Definition 2.2** (Own orientation)**.** The *own orientation* of a level-$m$ supertile is counterclockwise if $m$ is even and clockwise if $m$ is odd. We write $\omega_m = (-1)^m$.

Thus a child and its parent are always oriented oppositely. Each supertile has a distinguished rim vertex $A$, its *crossover corner*, defined recursively: for a tile, $A = v_0$; for a Mystic, $A$ is the vertex $v_0$ of its first tile $\mathrm{Tile}(1,1)$; for a level-$(m+1)$ supertile, $A$ is the crossover corner

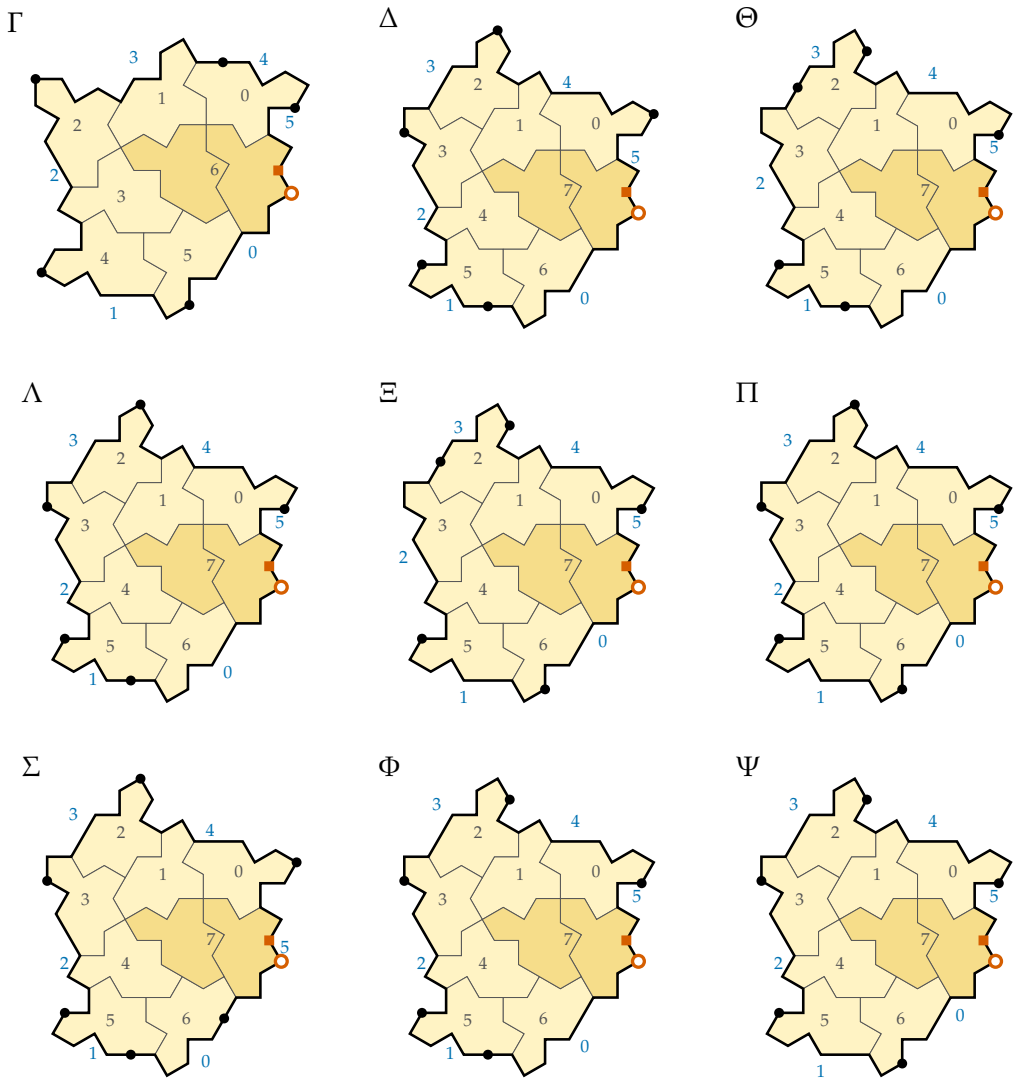



FIGURE 3. The nine level-1 supertiles, with true handedness. The Mystic is the darker pair of tiles. Grey numbers: the children (Table 1); black dots: the six vertices; blue numbers: the sides, numbered clockwise (the own orientation at level 1); open circle: the crossover corner $A$; filled square: the corner $B$ (Theorem 7.3). For Σ, $A$ lies inside side 5.

of its Γ child. (That this point lies on the rim is computed for levels 0 and 1 and follows from Proposition 2.4(d) at higher levels.) The second corner $B$, adjacent to $A$, appears in Theorem 7.3; we call $A$ and $B$ the *crossover corners*. The vertices of a level-$m$ supertile ($m \geq 1$) are numbered $0, \ldots, 5$ in own order, starting with the first vertex at or after $A$; side $j$ runs from vertex $j$ to vertex $j+1$. Figure 3 shows this numbering for the level-1 supertiles.

2.4. **The composition pattern.** Theorem 2.1(b) says that every level of the hierarchy is assembled in the same combinatorial way. We record this in our numbering.

**Definition 2.3** (Composition pattern)**.** The *composition pattern* $C$ consists of the following data for each parent kind $T$, given in Tables 1 to 3 and drawn in Fig. 4.

(i) The kinds $K_0, \ldots, K_{n_T-1}$ of the children (Table 1).
(ii) For each child $c$ and each side $j$ of $c$, either the sibling $c'$ and its side $j'$ that lie across side $j$, or the symbol $\partial$ if side $j$ lies on the parent's rim (Table 2).

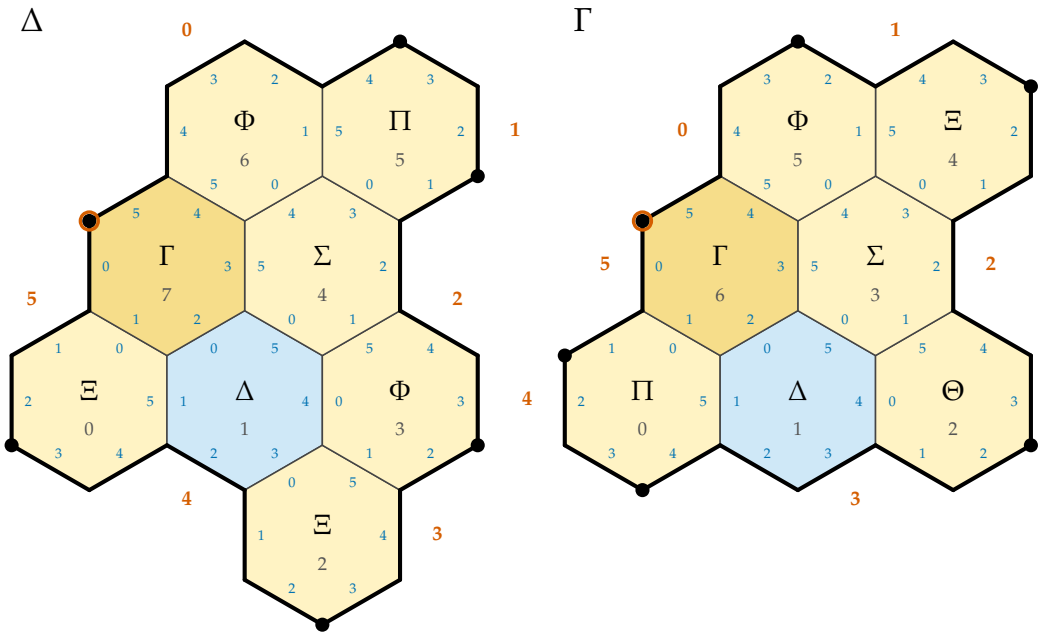



FIGURE 4. The composition pattern $C$ for $\Delta$ (left; the same for all kinds except $\Gamma$, with the children's kinds as in Table 1) and for $\Gamma$ (right), in the layout of [17, Fig. 5.1]. Each hexagon is a child: its kind, its number (grey), and its sides 0–5 (blue). Thick black: the parent's rim; black dots: the parent's vertices; vermilion numbers: the parent's sides; circled: the parent's crossover corner, which is vertex 0 of the $\Gamma$ child. Children's sides are numbered in one rotational sense and the parent's in the other.

(iii) For each parent side $J$, the sequence of child sides $(c, j)$ that make up side $J$, in the parent's own order (Table 3).

**Proposition 2.4** (The pattern holds at every level)**.** *Let $\mathcal{T}$ be a homochiral tiling by* Tile$(1,1)$*, let $m \geq 1$, and let $Y$ be a level-$(m+1)$ supertile of $\mathcal{T}$ of kind $T$, with children $X_0, \ldots, X_{n_T-1}$. Then:*

(a) *$X_c$ has kind $K_c$ (Table 1);*
(b) *side $j$ of $X_c$ is the same path of tile edges as side $j'$ of $X_{c'}$, traversed in the opposite direction, whenever Table 2 lists $c'.j'$, and it lies on the rim of $Y$ when the table lists $\partial$;*
(c) *side $J$ of $Y$ is the concatenation of the children's sides listed in Table 3, and vertex $J$ of $Y$ is the vertex $j+1$ of the child $c$ whose side $c.j$ is listed first;*
(d) *the crossover corner $A$ of $Y$ is vertex 0 of its $\Gamma$ child, which is vertex 0 of $Y$ if $T \neq \Sigma$, and lies inside side 5 of $Y$ if $T = \Sigma$.*

*Justification.* By Theorem 2.1(b), the level-$m$ supertiles inside $Y$ are combinatorially the hexagons of the corresponding panel of [17, Fig. 5.1]: hexagon edges correspond to sides and hexagon vertices to vertices. We use the following dictionary between the labels of [17, Fig. 4.2] and our side numbers:

(∗) side $j$ of a level-$m$ supertile of kind $K$ carries the label of the edge of hexagon $K$ in direction W, SW, SE, E, NE, NW for $j = 0, \ldots, 5$ respectively (Table 4); its crossover corner $A$ is vertex 0 if $K \neq \Sigma$; if $K = \Sigma$, $A$ is not a vertex, so it lies in the relative interior of side 5.

We show (∗) by induction on $m$. For $m = 1$ it is a computation on the level-1 supertiles, which have $E_1 = 44$ rim edges for $\Gamma$ and 46 for the other kinds; their vertices lie at the rim

| child | side 0 | side 1 | side 2 | side 3 | side 4 | side 5 |
|---|---|---|---|---|---|---|
| 0 | 7.1 | $\partial$ | $\partial$ | $\partial$ | $\partial$ | 1.1 |
| 1 | 7.2 | 0.5 | $\partial$ | 2.0 | 3.0 | 4.0 |
| 2 | 1.3 | $\partial$ | $\partial$ | $\partial$ | $\partial$ | 3.1 |
| 3 | 1.4 | 2.5 | $\partial$ | $\partial$ | $\partial$ | 4.1 |
| 4 | 1.5 | 3.5 | $\partial$ | 5.0 | 6.0 | 7.3 |
| 5 | 4.3 | $\partial$ | $\partial$ | $\partial$ | $\partial$ | 6.1 |
| 6 | 4.4 | 5.5 | $\partial$ | $\partial$ | $\partial$ | 7.4 |
| 7 | $\partial$ | 0.0 | 1.0 | 4.5 | 6.5 | $\partial$ |

| child | side 0 | side 1 | side 2 | side 3 | side 4 | side 5 |
|---|---|---|---|---|---|---|
| 0 | 6.1 | $\partial$ | $\partial$ | $\partial$ | $\partial$ | 1.1 |
| 1 | 6.2 | 0.5 | $\partial$ | $\partial$ | 2.0 | 3.0 |
| 2 | 1.4 | $\partial$ | $\partial$ | $\partial$ | $\partial$ | 3.1 |
| 3 | 1.5 | 2.5 | $\partial$ | 4.0 | 5.0 | 6.3 |
| 4 | 3.3 | $\partial$ | $\partial$ | $\partial$ | $\partial$ | 5.1 |
| 5 | 3.4 | 4.5 | $\partial$ | $\partial$ | $\partial$ | 6.4 |
| 6 | $\partial$ | 0.0 | 1.0 | 3.5 | 5.5 | $\partial$ |

TABLE 2. Adjacency of the children's sides: the entry $c'.j'$ in row $c$, column $j$ means that side $j$ of child $c$ is side $j'$ of child $c'$; $\partial$ means that it lies on the parent's rim. Left: every parent kind except $\Gamma$. Right: $\Gamma$.

*Eight children* (every $T \neq \Gamma$): the parent's rim, in own order from vertex 0 of child 7.

| position | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| piece | 7.5 | 6.4 | 6.3 | 6.2 | 5.4 | 5.3 | 5.2 | 5.1 | 4.2 | 3.4 | 3.3 |
| position | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| piece | 3.2 | 2.4 | 2.3 | 2.2 | 2.1 | 1.2 | 0.4 | 0.3 | 0.2 | 0.1 | 7.0 |

| parent | first position of side | | | | | |
|---|---|---|---|---|---|---|
| | 0 | 1 | 2 | 3 | 4 | 5 |
| $\Delta$ | 0 | 5 | 7 | 11 | 14 | 19 |
| $\Theta$ | 0 | 5 | 7 | 13 | 15 | 20 |
| $\Lambda$ | 0 | 5 | 7 | 11 | 14 | 20 |
| $\Xi$ | 0 | 3 | 7 | 13 | 15 | 20 |
| $\Pi$ | 0 | 3 | 7 | 11 | 14 | 20 |
| $\Sigma$ | 2 | 5 | 7 | 11 | 14 | 19 |
| $\Phi$ | 0 | 5 | 7 | 11 | 15 | 20 |
| $\Psi$ | 0 | 3 | 7 | 11 | 15 | 20 |

*Seven children* ($T = \Gamma$), from vertex 0 of child 6; sides $0, \ldots, 5$ begin at positions $0, 3, 6, 11, 16, 18$.

| position | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| piece | 6.5 | 5.4 | 5.3 | 5.2 | 4.4 | 4.3 | 4.2 | 4.1 | 3.2 | 2.4 |
| position | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 |
| piece | 2.3 | 2.2 | 2.1 | 1.3 | 1.2 | 0.4 | 0.3 | 0.2 | 0.1 | 6.0 |

TABLE 3. The parent's sides as concatenations of children's sides. A piece $c.j$ is side $j$ of child $c$, which the parent's own orientation traverses from vertex $j+1$ to vertex $j$ of the child. Side $J$ of the parent consists of the pieces from its first position up to the first position of side $J+1$ (cyclically). For $T \neq \Sigma$ the parent's side 0 begins at vertex 0 of the $\Gamma$ child, which is the parent's crossover corner. For $T = \Sigma$ the crossover corner lies inside side 5 (positions 19–1), after the pieces 0.2, 0.1, 7.0.

positions, counted from $A$ in own order,

$\Gamma : 0, 7, 14, 24, 34, 39$ $\quad \Delta : 0, 10, 15, 23, 30, 40$ $\quad \Theta : 0, 10, 15, 26, 31, 41$
$\Lambda : 0, 10, 15, 23, 30, 41$ $\quad \Xi : 0, 7, 15, 26, 31, 41$ $\quad \Pi : 0, 7, 15, 23, 30, 41$
$\Sigma : 4, 10, 15, 23, 30, 40$ $\quad \Phi : 0, 10, 15, 23, 31, 41$ $\quad \Psi : 0, 7, 15, 23, 31, 41.$

Drawn with the edges in odd directions lengthened to $\sqrt{3}$, which turns the tiles into hats and turtles [17, §3], and in the handedness of [17, Fig. 4.1], the nine level-1 supertiles coincide with the nine marked clusters there: the same outlines, the vertices at the marked dots, and the labels of Table 4 in the places where that figure has them; $A$ is the dot between the sides labelled W and NW (for $\Sigma$, a point of the NW side), on the light tile of the Mystic. A script checks this against a render of the figure (Appendix A): for each cluster it finds the six dots, fits our six vertices to them by a proper similarity with residual below 1% of an edge (any other matching, or the vertices of any other kind, fits at least 40 times worse), finds our whole rim on the printed outline and our first Mystic tile on the lighter shaded tile, and compares the labels, read from the figure by hand, with Table 4. Suppose $(*)$ holds for the children of $Y$. The panel does not label the inner hexagons, but the matching rules fix the rotation of each of them, and there is exactly one consistent choice (it is found by the script of Remark 2.5). With this dictionary, Tables 2 and 3 are then a transcription of the panel of [17, Fig. 5.1] for $T$, including its vertices (the six marked points on its outline); this gives (a)–(c). The crossover corner of $Y$ is that of its $\Gamma$ child, which is the child's vertex 0 by $(*)$, and Table 3 places this vertex at the start of the parent's side 0 if $T \neq \Sigma$ and inside side 5 if $T = \Sigma$; this is (d). Finally, the labels that the panel assigns to the parent's sides, numbered from this point in the parent's own order, are those of Table 4 for every $T$. So $(*)$ holds for $Y$. In particular $A$ always lies on the rim. □

*Remark* 2.5 (Computer cross-checks of the pattern). Since Tables 2 and 3 carry the whole weight of the citation, we checked them in two independent ways. First, we built the actual supertiles of levels 1 to 8 with the key-point construction of [17, App. A] and extracted the pattern of each composition $m \to m+1$ for $m = 1, \dots, 7$. For $m \leq 4$ the vertices were found as the points where three level-$m$ supertiles meet, observed inside larger supertiles. For $m = 5, \dots, 8$ they were propagated by the rule of Proposition 2.4(c), learnt at the observed level of the same parity; at levels 5 and 6 the propagated vertices agree with the observed ones (for $\Psi$, with the four of its six vertices that were observed). So for $m = 6, 7$ the rule (c) is an input of the extraction rather than something it tests. All seven extracted patterns are identical to each other and to Tables 2 and 3; for $m \leq 3$ this was done in exact arithmetic (Appendix A). Second, the nine panels of [17, Fig. 5.1] were encoded: kinds, positions and edge labels by hand, and the marked vertices (with the centre of each panel's $\Gamma$ hexagon) by a script from a render of the figure, which re-runs from the PDF. A script then confirmed, for all nine kinds, that the encoding agrees with Tables 2 and 3, and that the side-to-label dictionary above is the same in all 71 placements of a child. Two deliberately corrupted encodings (two outer labels exchanged; one wrong label in [17, Fig. 4.2]) were rejected. We also checked that the tiles of every supertile of levels 1 to 7 (up to 2,147,679 tiles) have disjoint interiors and exactly fill its rim.

We shall use one consequence of Theorem 2.1(a) and Proposition 2.4 repeatedly.

**Lemma 2.6** (Rigidity)**.** *Any two level-$m$ supertiles of the same kind, in possibly different homochiral tilings, are related by a rotation and translation that maps descendants to descendants with their numbers (children, their children, and so on, down to the tiles), and it is the only such motion. For $m \geq 1$ it maps the six vertices, the sides and the corner $A$ of one onto those of the other.*

*Proof.* Induction on $m$. For $m = 0$ this is clear, since the Mystic is a fixed pair of tiles without proper symmetry. For $m = 1$, the level-1 supertiles of a homochiral tiling are congruent copies of the clusters of Fig. 3 [17, Fig. 2.1, §4], with their vertices at the points marked in [17, Fig. 4.1]; the numbers of their children are those of the reference clusters of Fig. 3, and $A$ is fixed by the Mystic. Let $m \geq 1$. By induction, the children of two level-$(m+1)$ supertiles $Y, Y'$ of kind $T$ are pairwise related by unique proper motions $g_c$, which map the vertices of child $c$ of $Y$ onto those of child $c$ of $Y'$ with their numbers. If side $j$ of child $c$ is glued to side $j'$ of child $c'$, then by Proposition 2.4(b), applied to $Y$ and to $Y'$, vertices $j$ and $j+1$ of $c$ coincide with vertices $j'+1$ and $j'$ of $c'$, in $Y$ and in $Y'$. So $g_c$ and $g_{c'}$ agree on two distinct

points; being proper motions, they are equal. The children's adjacency graph is connected, so all the $g_c$ are one motion $g$. By Proposition 2.4(c),(d), the vertices and the corner $A$ of $Y$ are vertices and the corner $A$ of children, so $g$ maps them, and hence the sides, onto those of $Y'$. Since $g$ agrees with each $g_c$, it maps descendants to descendants with their numbers. Any proper motion that does so maps the first tile of the Mystic at the bottom of the chain of last ($\Gamma$) children to its counterpart, so it equals $g$, since a tile has no symmetry. □

Write $E_m(K)$ for the number of rim edges (equally, rim vertices) of a level-$m$ supertile of kind $K$, and $\ell_m(K, j)$ for the number of edges of its side $j$. By Lemma 2.6 these numbers are well defined.

## 3. The decoration and its strands

3.1. **Ports and pieces.** Let $t$ be a tile and $e$ one of its edges, with endpoints $x$ and $y$. The tile has two *ports* on $e$, the port *near* $x$ and the port *near* $y$; formally, they are the pairs $(e, x)$ and $(e, y)$. In the drawing (Fig. 5), $e$ carries one crossing point, and the two ribbon ends of $t$ that meet it, one running off towards $x$ and one towards $y$, are these two ports. For a vertex $v_i$ of the tile, write

$$\mathrm{R}(v_i) = \text{the port on } e_i \text{ near } v_i, \qquad \mathrm{L}(v_i) = \text{the port on } e_{i-1} \text{ near } v_i,$$

the ports just after and just before the corner $v_i$ in counter-clockwise order. Each of the 28 ports is $\mathrm{R}(v_i)$ or $\mathrm{L}(v_i)$ for exactly one $i$.

**Definition 3.1** (The decoration)**.** The *decoration* joins the 28 ports of Tile(1, 1) in pairs by fourteen *pieces*:

$$\underbrace{\mathrm{L}(v_i) \leftrightarrow \mathrm{R}(v_i) \quad (2 \le i \le 13)}_{\text{twelve hooks}}, \qquad \underbrace{\mathrm{R}(v_0) \leftrightarrow \mathrm{R}(v_1), \qquad \mathrm{L}(v_0) \leftrightarrow \mathrm{L}(v_1)}_{\text{the crossover}}.$$

Every tile of a homochiral tiling carries the image of this decoration.

In our drawing the two crossover pieces cross exactly once, transversally, and this is the only crossing inside the tile (Remark 3.6). Each hook turns back round one vertex. Figure 5 shows the pieces and their ports, and Fig. 1 the ribbons as drawn. The drawing uses cubic Bézier curves, with the crossing point of edge $e_i$ at distance $0.5 + 0.15(-1)^i$ from $v_i$. Only the pairing of Definition 3.1 matters for the theorems; any drawing that realises it may be used.

3.2. **The straight-through rule.** Let $\mathcal{T}$ be a homochiral tiling, and let $e$ be an edge shared by tiles $t$ and $t'$, with endpoints $x$ and $y$.

**Definition 3.2** (Straight-through rule)**.** The port of $t$ on $e$ near $x$ is *joined* to the port of $t'$ on $e$ near $y$, and the port of $t$ near $y$ to the port of $t'$ near $x$.

This is what a ribbon does when it continues straight through the crossing point (Fig. 6). In the drawing the two tiles' crossing points on $e$ coincide. The reason is that tiles in a homochiral tiling traverse a shared edge in opposite directions, and an edge $e_i$ of one tile is always glued to an edge $e_j$ of the other with $i + j$ odd; this follows from the vertex colouring of [17, proof of Lemma 2.1]: the interior angles of Tile(1, 1) alternate between multiples of 90° and of 120°, and these classes must agree where vertices meet. As $v_i$ of one tile meets $v_{j+1}$ of the other, $i$ and $j + 1$ have the same parity. We verified it on all supertiles up to level 4. We also checked that the drawn ribbons realise the pairing on all supertiles up to level 3 (Appendix A). The proofs use only Definitions 3.1 and 3.2.

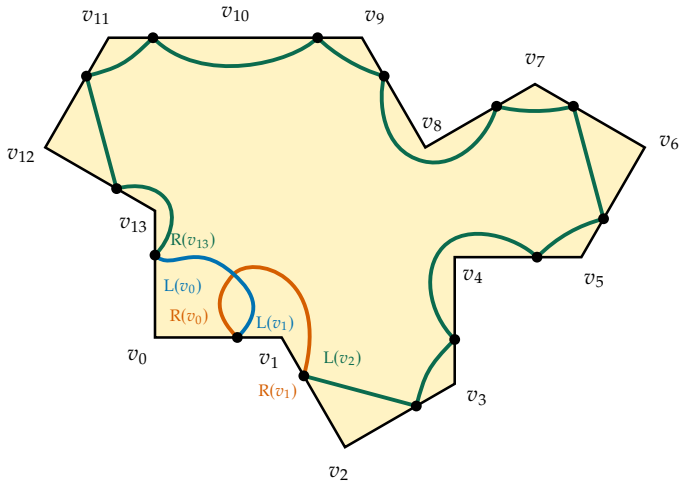



FIGURE 5. The decoration on the true geometry (centre lines of the ribbons of Fig. 1). Each edge carries one crossing point (dot), at which two pieces of the tile end: one leaves towards each end of the edge. On $e_i$ these two ends are the ports $\mathrm{R}(v_i)$ (towards $v_i$) and $\mathrm{L}(v_{i+1})$ (towards $v_{i+1}$); the labels show them round the crossover corners. Green: the twelve hooks $\mathrm{L}(v_i) \leftrightarrow \mathrm{R}(v_i)$. Vermilion: $\mathrm{R}(v_0) \leftrightarrow \mathrm{R}(v_1)$; blue: $\mathrm{L}(v_0) \leftrightarrow \mathrm{L}(v_1)$, the crossover at the corners $A = v_0$ and $B = v_1$.

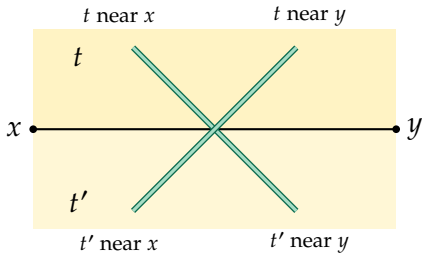



FIGURE 6. The straight-through rule at the crossing point of an edge $xy$: the end of $t$ heading towards $x$ continues as the end of $t'$ heading towards $y$.

### 3.3. **Strands.**

**Definition 3.3** (Strand graph)**.** Let $\mathcal{T}$ be a homochiral tiling. The *strand graph* $\mathcal{G}(\mathcal{T})$ is the multigraph with one vertex for each piece of each tile and one edge for each pair of ports joined by the straight-through rule, between the two pieces having these ports as ends. A *strand* is a connected component of $\mathcal{G}(\mathcal{T})$; a strand is *closed* if it is finite.

**Lemma 3.4.** *Every vertex of $\mathcal{G}(\mathcal{T})$ has degree 2. Hence every strand is either a finite cycle (a closed strand) or a two-way infinite path.*

*Proof.* A piece has two ends, each a port of its tile on some edge $e$. In a tiling of the plane, $e$ is shared by exactly one other tile, so each end is joined to exactly one port, an end of a piece of that neighbour. Joins always link pieces of different tiles, so $\mathcal{G}(\mathcal{T})$ has no loops, and every vertex has degree exactly 2. A connected, locally finite multigraph with all degrees 2 is a finite cycle (possibly of length 2) or a two-way infinite path. □

*Remark* 3.5 (A single strand needs a non-periodic tiling). Let $\mathcal{T}'$ be any edge-to-edge tiling of the plane by compact decorated tiles (not necessarily copies of Tile(1, 1)), each carrying

at least one piece, with one crossing point on each edge, and with strands defined as above by the straight-through rule. Suppose that $\mathcal{T}'$, decoration included, is invariant under a translation $\tau \neq 0$, and that its strand graph is a single two-way infinite path, with vertices $x \in \mathbb{Z}$. Then $\tau$ acts on the path by a graph automorphism, and freely, since a non-zero translation moves every tile. An automorphism of the path is $x \mapsto x + k$ or $x \mapsto c - x$. A reflection $x \mapsto c - x$ satisfies $\tau^2 = \mathrm{id}$ on the path, but $\tau^2$ is again a non-zero translation and fixes no piece; so $\tau$ is $x \mapsto x + k$ with $k \neq 0$, and the pieces fall into $|k|$ orbits under $\langle\tau\rangle$. But the tiles fall into infinitely many orbits: the images of one tile from each orbit cover the cylinder $\mathbb{R}^2/\langle\tau\rangle$, which has infinite area. Since every tile has a piece, the pieces also fall into infinitely many orbits. This contradiction shows that the ribbons of a decorated tiling with a translational symmetry never form a single strand. Every tiling by Tile$(1, 1)$ of one handedness is non-periodic [17, Thm. 2.2], so this obstruction does not arise there.

*Remark* 3.6 (Over and under). Draw the pieces of every tile as in Fig. 1. We checked numerically that each piece stays inside its tile away from its ends, that the two crossover pieces cross exactly once, transversally (at about 90°), and that no other two pieces of a tile meet except at their common ends on the edges; and that at each crossing point the two pieces of a tile meet the edge at ±45° (Appendix A). So a ribbon continues smoothly across an edge, and the curves in the plane cross only transversally, in pairs: once inside each tile and at right angles at each edge. Colour each region of the complement of the curves by the parity of the number of crossings of a generic path from a fixed region. This is well defined: the parity of the number of crossings of a generic closed loop does not change when the loop is moved across an arc of the curves, or across a crossing point, where four arcs meet; and every closed loop in the plane can be shrunk to a point. Adjacent regions get different colours. The usual checkerboard rule then makes over and under alternate along every strand [6]. With such a choice the single strand of the Ariadne Theorem becomes an alternating diagram of a line, as in Celtic knotwork. None of this affects which pieces lie on which strand.

## 4. Boxes and the signature

4.1. **Patches and boxes.** A *patch* is a finite set $P$ of tiles of a homochiral tiling whose union is a closed disc with a simple polygonal boundary, the *rim* $\partial P$. Every supertile of level $\geq 1$ is a patch (Theorem 2.1), and so are a tile and a Mystic (whose two tiles share four consecutive edges, $e_6, \ldots, e_9$ of the first tile; its rim has 20 edges). Every rim edge belongs to exactly one tile of $P$. Let $E = E(P)$ be the number of rim edges, and number the rim vertices $u_0, \ldots, u_{E-1}$ in a chosen orientation of the rim, indices mod $E$. For a supertile we always use its own orientation (Definition 2.2). For a rim vertex $u_k$ define the *rim ports*

$$\mathrm{R}(u_k) = \text{the port on } u_k u_{k+1} \text{ near } u_k, \qquad \mathrm{L}(u_k) = \text{the port on } u_{k-1} u_k \text{ near } u_k,$$

where the port belongs to the tile of $P$ containing that edge. There are $2E$ rim ports. For the tile itself, with its counter-clockwise orientation, these are the ports $\mathrm{R}(v_i)$, $\mathrm{L}(v_i)$ of Section 3. Reversing the orientation exchanges R and L at every vertex.

**Lemma 4.1** (The box of a patch)**.** *Start at a rim port of $P$ and follow the pieces of the tiles of $P$, crossing every interior edge of $P$ by the straight-through rule. The path ends at a different rim port. This defines an involution $\pi_P$ on the rim ports without fixed points. The pieces of $P$ that are not reached from any rim port form finitely many cycles, the* closed strands *of $P$; write $\kappa(P)$ for their number.*

*Proof.* The pieces of $P$ and the joins across interior edges form a finite graph in which every piece has degree 2, except that a piece end at a rim port has no partner. Its components are therefore paths, whose two ends are rim ports, and cycles. Paths have two distinct ends. □

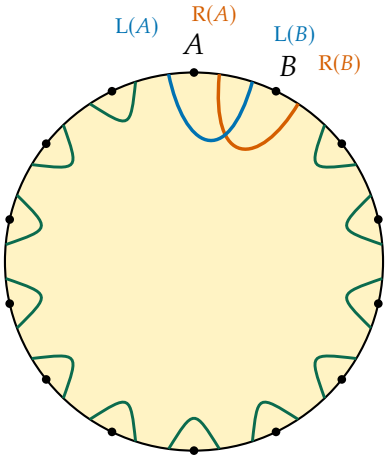



FIGURE 7. The signature at $(A, B)$, drawn for a rim with 14 vertices oriented clockwise: a hook at every rim vertex except $A$ and $B$, and the two crossover arcs $\mathrm{R}(A) \leftrightarrow \mathrm{R}(B)$ and $\mathrm{L}(A) \leftrightarrow \mathrm{L}(B)$.

We call the paths of Lemma 4.1 the *arcs* of $P$; there are exactly $E$ of them. The triple $(\partial P, \pi_P, \kappa(P))$ is the *box* of $P$. Everything we need about the inside of $P$ is in its box, as the next lemma shows.

**Lemma 4.2** (Gluing)**.** *Let a patch $P$ be the union of patches $P_1, \dots, P_n$ with pairwise disjoint interiors. Then the box of $P$ is determined by the boxes of the $P_i$ together with the identification of their shared rim edges. More precisely, $\pi_P$ is computed by alternately applying the $\pi_{P_i}$ and the straight-through rule across shared edges, until a rim port of $P$ is reached. The closed strands of $P$ are those of the $P_i$ together with the cycles of this alternation that contain no rim port of $P$.*

*Proof.* Every arc or closed strand of $P$ is a concatenation of arcs and closed strands of the $P_i$, joined across edges shared by two of the $P_i$. □

4.2. **The signature.**

**Definition 4.3** (Signature)**.** Let $A$ and $B$ be adjacent rim vertices of a patch $P$. We say that $P$ has the *signature at* $(A, B)$ if

(i) $\pi_P(\mathrm{L}(u)) = \mathrm{R}(u)$ for every rim vertex $u \notin \{A, B\}$ (a *hook* at $u$);
(ii) $\pi_P(\mathrm{R}(A)) = \mathrm{R}(B)$ and $\pi_P(\mathrm{L}(A)) = \mathrm{L}(B)$ (the *crossover*);
(iii) $\kappa(P) = 0$.

The definition is symmetric in $A$ and $B$ and does not depend on the orientation of the rim, since reversing it exchanges L and R. A patch with the signature has $E - 2$ hook arcs, one at each rim vertex other than $A$ and $B$, and two crossover arcs. Figure 7 is a picture. By Definition 3.1:

**Lemma 4.4.** *A tile has the signature at $(v_0, v_1)$, and a Mystic has the signature at the vertices $v_0, v_1$ of its first tile.*

*Proof.* For the tile this is Definition 3.1. For the Mystic it is a finite computation with Lemma 4.2: there is no closed strand, the 18 rim vertices other than these two are hooks, and the crossover persists (Appendix A). □

## 5. The zigzag lemma and the skeleton

Throughout this section a patch $P$ is the union of patches (*children*) $X_0, \dots, X_{n-1}$ with disjoint interiors, all of them oriented in the same rotational sense, and $P$ is oriented in that sense too. Each child $X_c$ has the signature at $(A_c, B_c)$. When $P$ is a supertile and the

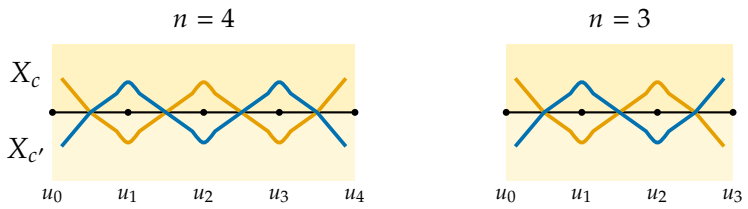



FIGURE 8. The zigzag lemma. Along a shared path whose interior corners are hooks on both sides, the ribbons cross the path at every edge and turn back at every corner, alternately in $X_c$ (above) and $X_{c'}$ (below). For even $n$ each connection returns to the side it started from; for odd $n$ it ends on the other side.

$X_c$ are its children, the children carry their own orientation, so $P$ is oriented opposite to its own orientation. This does not matter for the signature, which does not depend on the orientation (Section 4); but it does exchange L and R at the parent's rim vertices.

5.1. **Runs and the zigzag lemma.** Two children $c \neq c'$ oriented in the same sense traverse a common rim path in opposite directions. For a rim vertex $u$ of $X_c$ we write $\mathrm{R}_c(u), \mathrm{L}_c(u)$ for the rim ports of $X_c$ in the common orientation fixed above (for supertiles, their own orientation).

**Lemma 5.1** (Zigzag Lemma)**.** *Let $u_0u_1 \cdots u_n$ ($n \geq 1$) be a path of $n$ edges lying on the rims of both $X_c$ and $X_{c'}$, traversed by $X_c$ from $u_0$ to $u_n$. Suppose that none of the interior vertices $u_1, \ldots, u_{n-1}$ is a crossover corner of $X_c$ or of $X_{c'}$. Then, alternately applying the straight-through rule across the edges $u_ku_{k+1}$ and the hooks of $X_c$ and $X_{c'}$ at $u_1, \ldots, u_{n-1}$, the ports at the two ends of the path are joined as follows:*

$$\begin{aligned} &n \text{ even:} && \mathrm{R}_c(u_0) \leftrightarrow \mathrm{L}_c(u_n), && \mathrm{R}_{c'}(u_n) \leftrightarrow \mathrm{L}_{c'}(u_0); \\ &n \text{ odd:} && \mathrm{R}_c(u_0) \leftrightarrow \mathrm{R}_{c'}(u_n), && \mathrm{L}_c(u_n) \leftrightarrow \mathrm{L}_{c'}(u_0). \end{aligned}$$

*These two connections use every port on the edges of the path, so no closed strand lies along the path. Moreover, for $0 < k < n$, the hook of $X_c$ at $u_k$ lies on the connection through $\mathrm{R}_c(u_0)$ if $k$ is even. If $k$ is odd, it lies on the connection through $\mathrm{L}_c(u_n)$ when $n$ is odd, and on the connection through $\mathrm{R}_{c'}(u_n)$ when $n$ is even.*

*Proof.* Start at $\mathrm{R}_c(u_0)$, the port of $X_c$ on $u_0u_1$ near $u_0$. The straight-through rule takes it to the port of $X_{c'}$ on $u_0u_1$ near $u_1$. In the orientation of $X_{c'}$, which runs from $u_1$ to $u_0$ along this edge, that port is $\mathrm{R}_{c'}(u_1)$. If $n = 1$ we have reached $\mathrm{R}_{c'}(u_n)$. Otherwise $u_1$ is a hook of $X_{c'}$, which leads to $\mathrm{L}_{c'}(u_1)$, the port of $X_{c'}$ on $u_1u_2$ near $u_1$. Crossing $u_1u_2$ leads to the port of $X_c$ on $u_1u_2$ near $u_2$, which is $\mathrm{L}_c(u_2)$. If $n = 2$ we stop. Otherwise the hook of $X_c$ at $u_2$ leads to $\mathrm{R}_c(u_2)$, and the pattern repeats with all indices shifted by two. So the path visits the hooks of $X_{c'}$ at odd $k$ and the hooks of $X_c$ at even $k$. It ends at $\mathrm{L}_c(u_n)$ if $n$ is even and at $\mathrm{R}_{c'}(u_n)$ if $n$ is odd. The same argument from the other end, starting at $\mathrm{R}_{c'}(u_n)$ for even $n$ and at $\mathrm{L}_c(u_n)$ for odd $n$, gives the second connection; it visits the remaining hooks. Each connection crosses each of the $n$ edges once, using two of its four ports, so together they use all of them. □

5.2. **Marks, runs and the skeleton.**

**Definition 5.2** (Marks and runs)**.** A *mark* is a point that is a rim vertex of some child and is either

(i) a *junction*: a rim vertex of a child $X_c$ at which the two rim edges of $X_c$ have different neighbours, where a neighbour is a sibling or "outside $P$"; or
(ii) one of the crossover corners $A_c$, $B_c$ of some child.

A *run* of $X_c$ is a path along the rim of $X_c$ between two consecutive marks lying on that rim. (In every composition below each child has at least four marks on its rim, so runs are arcs with distinct ends.)

Along a run the neighbour of $X_c$ does not change. So a run either lies on the rim of $P$ (a *rim run*) or is shared with exactly one sibling $X_{c'}$ (an *interior run*). In the second case it is also a run of $X_{c'}$, its *partner*, traversed in the opposite direction, because marks are points and the same marks bound it on both sides. At an interior vertex $u$ of an interior run, $X_c$ and $X_{c'}$ share both edges at $u$, so the full angle at $u$ is filled by these two children and no third child touches $u$. Since crossover corners are marks, the zigzag lemma applies to every interior run.

**Definition 5.3** (Skeleton)**.** The *skeleton* $\mathcal{S}$ of the composition $P = \bigcup X_c$ consists of the following data: the children and their kinds; for each child, the cyclic sequence of its marks and runs, and which of its marks are $A_c$ and $B_c$; for each interior run, its partner run and the parity of its length; the cyclic order of the rim runs round $\partial P$.

Two skeletons of compositions with the same children (by number and kind) are *equal* if there is a bijection of their marks that preserves, for each child, the cyclic order of its marks and the labels $A_c, B_c$, and preserves partner runs, the parities of interior runs and the cyclic order of the rim runs. The maps $\tau$ and $\sigma$ and the parent ports below are defined from these data alone, so such a bijection carries them, and hence the conclusions of Proposition 5.4, from one composition to the other.

The *reduced ports* are the ports $\mathrm{R}_c(u)$ and $\mathrm{L}_c(u)$ for every child $c$ and every mark $u$ on its rim. The port $\mathrm{R}_c(u)$ lies on the run of $X_c$ that starts at $u$, and $\mathrm{L}_c(u)$ on the run that ends at $u$. Define two involutions on reduced ports.

- *Inside a child*: $\tau(\mathrm{L}_c(u)) = \mathrm{R}_c(u)$ if $u \notin \{A_c, B_c\}$; $\tau(\mathrm{R}_c(A_c)) = \mathrm{R}_c(B_c)$ and $\tau(\mathrm{L}_c(A_c)) = \mathrm{L}_c(B_c)$.
- *Across an interior run* from $u$ to $u'$ of $X_c$, of length $n$, with partner in $X_{c'}$: if $n$ is even, $\sigma(\mathrm{R}_c(u)) = \mathrm{L}_c(u')$ and $\sigma(\mathrm{R}_{c'}(u')) = \mathrm{L}_{c'}(u)$; if $n$ is odd, $\sigma(\mathrm{R}_c(u)) = \mathrm{R}_{c'}(u')$ and $\sigma(\mathrm{L}_c(u')) = \mathrm{L}_{c'}(u)$.

Both $\tau$ and $\sigma$ are determined by $\mathcal{S}$. The *parent ports* are the reduced ports on rim runs. They are exactly the rim ports $\mathrm{R}(u), \mathrm{L}(u)$ of $P$ at the rim vertices $u$ of $P$ that are marks (the *parent marks*), since $X_c$ and $P$ induce the same orientation on a rim run.

**Proposition 5.4** (Reduced composition)**.** *Let $P = \bigcup X_c$, with every child having the signature. Then:*

(a) *every rim vertex of $P$ that is not a mark is a hook of $P$;*
(b) *for a parent port $p$, $\pi_P(p)$ is the first parent port among $\tau(p)$, $\tau\sigma\tau(p)$, $\tau\sigma\tau\sigma\tau(p), \ldots$ (the ports $\sigma\tau(p), \ldots$ in between lie on interior runs);*
(c) *$\kappa(P)$ is the number of cycles of the alternation of $\sigma$ and $\tau$ that contain no parent port.*

*In particular, $\pi_P$ at the parent marks and $\kappa(P)$ are determined by the skeleton.*

*Proof.* (a) Let $u$ be a rim vertex of $P$ that is not a mark. The rim edge of $P$ after $u$ lies in some $X_c$; if the other rim edge of $X_c$ at $u$ were shared with a sibling, $u$ would be a junction. So both rim edges of $P$ at $u$ belong to $X_c$, and $u$ lies inside a rim run of $X_c$ and is not a crossover corner of $X_c$. Hence $\pi_{X_c}(\mathrm{L}_c(u)) = \mathrm{R}_c(u)$, and both ports are rim ports of $P$. (b) and (c) follow from Lemma 4.2: the ribbon starting at $p$ runs through $X_c$ to a reduced port $\tau(p)$, since by the signature of $X_c$ the arc of $X_c$ starting at a port at a mark ends at a port at a mark. If $\tau(p)$ lies on an interior run, the ribbon then follows that run, by the zigzag lemma, to $\sigma\tau(p)$. A closed strand of $P$ not inside a child must cross some shared edge, and by the zigzag lemma it then passes through a reduced port; following it from there gives a cycle of the alternation. Conversely each such cycle, read as arcs of children joined by zigzag connections, is a closed strand of $P$, and distinct cycles give distinct strands. The children have no closed strands. □

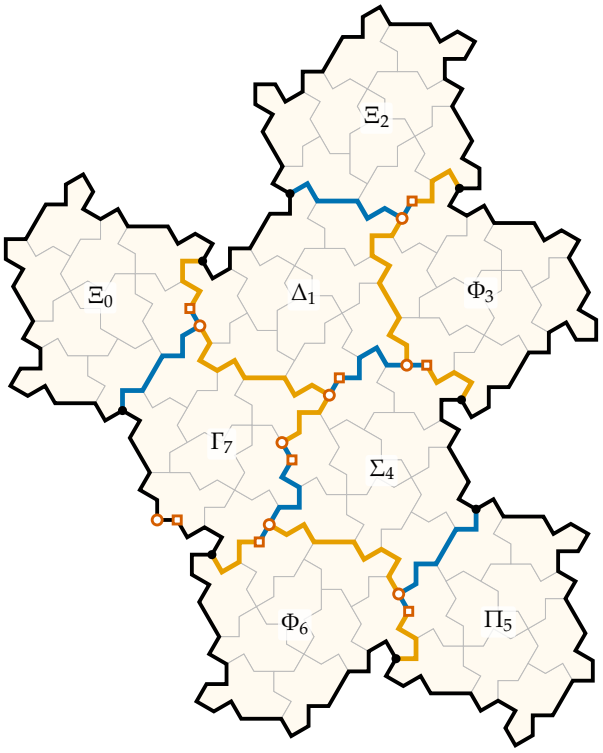



FIGURE 9. Marks and runs for the composition of a level-2 Δ-supertile from its eight level-1 children (subscripts are child numbers). Dots: the 24 marks (open circles and squares: the children's crossover corners $A_c$ and $B_c$). Shared runs are coloured by the parity of their length: blue odd, orange even. Black: rim runs. The skeleton records this picture up to the lengths of the runs, keeping only the parities of the shared runs.

**Corollary 5.5** (Where a hook goes)**.** *In the situation of Proposition 5.4, let $u$ be a rim vertex of $X_c$ that is not a mark. If $u$ lies inside a rim run, the hook of $X_c$ at $u$ is a hook of $P$. If $u$ lies inside an interior run, traversed by $X_c$ from $u_0$ to $u_n$, at distance $k$ from $u_0$, then the hook of $X_c$ at $u$ lies on the same arc or closed strand of $P$ as the reduced port $\mathrm{R}_c(u_0)$ if $k$ is even; if $k$ is odd, as $\mathrm{L}_c(u_n)$ when $n$ is odd and as $\mathrm{R}_{c'}(u_n)$ when $n$ is even.*

*Proof.* This is the last part of Lemma 5.1, together with Proposition 5.4(a). □

## 6. Side lengths

6.1. **Letters.** Every side of a level-$m$ supertile ($m \geq 1$) carries one of the edge labels of [17, Fig. 4.2] (Table 4). Ignoring the signs, there are eight *letters*, $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta$. Two sides that are glued to each other carry labels $x^+$ and $x^-$ (or $\eta$ and $\eta$), and so the same letter; this is the matching rule of [17, §4], and it can be read off Tables 1, 2 and 4.

**Proposition 6.1** (Lengths by letter)**.** *For every $m \geq 1$ the length $\ell_m(K, j)$ depends only on the letter of side $j$ of $K$. Writing $x_m$ for the common length of the sides with letter $x$,*

$$
\begin{aligned}
\alpha_{m+1} &= \beta_m + \epsilon_m + \eta_m, & \beta_{m+1} &= \alpha_m + \epsilon_m, \\
\gamma_{m+1} &= \beta_m + 2\epsilon_m + \eta_m + \theta_m, & \delta_{m+1} &= \alpha_m + \beta_m + \epsilon_m + 2\theta_m, \\
\epsilon_{m+1} &= 3\epsilon_m + \eta_m, & \zeta_{m+1} &= \alpha_m + \epsilon_m + \theta_m, \\
\eta_{m+1} &= 4\epsilon_m + \eta_m, & \theta_{m+1} &= 4\epsilon_m + \eta_m + \theta_m,
\end{aligned}
\tag{1}
$$

*with* $(\alpha_1, \beta_1, \gamma_1, \delta_1, \epsilon_1, \zeta_1, \eta_1, \theta_1) = (7, 5, 10, 10, 8, 6, 10, 11)$.

| kind | side 0 | side 1 | side 2 | side 3 | side 4 | side 5 |
|---|---|---|---|---|---|---|
| $\Gamma$ | $\alpha^-$ | $\alpha^+$ | $\gamma^-$ | $\delta^-$ | $\beta^+$ | $\beta^-$ |
| $\Delta$ | $\gamma^+$ | $\beta^+$ | $\epsilon^-$ | $\alpha^+$ | $\gamma^-$ | $\zeta^-$ |
| $\Theta$ | $\gamma^+$ | $\beta^+$ | $\theta^+$ | $\beta^+$ | $\eta$ | $\beta^-$ |
| $\Lambda$ | $\gamma^+$ | $\beta^+$ | $\epsilon^-$ | $\alpha^+$ | $\theta^-$ | $\beta^-$ |
| $\Xi$ | $\alpha^-$ | $\epsilon^+$ | $\theta^+$ | $\beta^+$ | $\eta$ | $\beta^-$ |
| $\Pi$ | $\alpha^-$ | $\epsilon^+$ | $\epsilon^-$ | $\alpha^+$ | $\theta^-$ | $\beta^-$ |
| $\Sigma$ | $\zeta^+$ | $\beta^+$ | $\epsilon^-$ | $\alpha^+$ | $\gamma^-$ | $\delta^+$ |
| $\Phi$ | $\gamma^+$ | $\beta^+$ | $\epsilon^-$ | $\epsilon^+$ | $\eta$ | $\beta^-$ |
| $\Psi$ | $\alpha^-$ | $\epsilon^+$ | $\epsilon^-$ | $\epsilon^+$ | $\eta$ | $\beta^-$ |

TABLE 4. The label of [17, Fig. 4.2] carried by side $j$ of a supertile of each kind: the label of the edge of the hexagon in direction W, SW, SE, E, NE, NW for $j = 0, \dots, 5$.

*Proof.* For $m = 1$ this is a direct count on the level-1 supertiles (Fig. 3; Appendix A). By Proposition 2.4(c), $\ell_{m+1}(T, J)$ is the sum of the lengths of the children's sides listed in Table 3. Inspection of Tables 1, 3 and 4 shows that the multiset of letters of these children's sides depends only on the letter of side $J$ of $T$, and is the one in (1). For example, side 0 of $\Delta$ (letter $\gamma$) is made of 7.5, 6.4, 6.3, 6.2, 5.4, with letters $\beta, \eta, \epsilon, \epsilon, \theta$. The claim follows by induction on $m$. □

The recurrence (1) has a surprisingly simple solution. Let $F_n$ be the Fibonacci numbers, $F_1 = F_2 = 1$.

**Proposition 6.2** (Fibonacci side lengths)**.** *For every $m \geq 1$,*

$$\alpha_m = F_{3m+2} + 2, \qquad \beta_m = F_{3m+1} + 2, \qquad \gamma_m = \delta_m = \eta_m = 2F_{3m+2},$$
$$\epsilon_m = F_{3m+3}, \qquad \zeta_m = 2F_{3m+1}, \qquad \theta_m = F_{3m+4} - 2.$$

*Proof.* For $m = 1$, $(F_4, F_5, F_6, F_7) = (3, 5, 8, 13)$ gives the initial values. For the induction step put $a = F_{3m+1}$ and $b = F_{3m+2}$, so that $F_{3m+3} = a + b$, $F_{3m+4} = a + 2b$, $F_{3m+5} = 2a + 3b$, $F_{3m+6} = 3a + 5b$ and $F_{3m+7} = 5a + 8b$. Substituting into (1):

$$\alpha_{m+1} = (a+2) + (a+b) + 2b = F_{3m+5} + 2, \qquad \beta_{m+1} = (b+2) + (a+b) = F_{3m+4} + 2,$$
$$\gamma_{m+1} = (a+2) + 2(a+b) + 2b + (a+2b-2) = 4a + 6b = 2F_{3m+5},$$
$$\delta_{m+1} = (b+2) + (a+2) + (a+b) + 2(a+2b-2) = 4a + 6b,$$
$$\eta_{m+1} = 4(a+b) + 2b = 4a + 6b, \qquad \epsilon_{m+1} = 3(a+b) + 2b = F_{3m+6},$$
$$\zeta_{m+1} = (b+2) + (a+b) + (a+2b-2) = 2F_{3m+4},$$
$$\theta_{m+1} = 4(a+b) + 2b + (a+2b-2) = 5a + 8b - 2 = F_{3m+7} - 2.$$ □

Since $F_n$ is even exactly when $3 \mid n$:

**Corollary 6.3** (Parities are constant)**.** *For every $m \geq 1$ the lengths $\alpha_m$, $\beta_m$, $\theta_m$ are odd and $\gamma_m$, $\delta_m$, $\epsilon_m$, $\zeta_m$, $\eta_m$ are even.*

For example, the rim of a level-$m$ $\Delta$-supertile has length $E_m(\Delta) = \alpha_m + \beta_m + 2\gamma_m + \epsilon_m + \zeta_m$, which is $46, 182, 758, 3198, 13534, \dots$.

### 6.2. **The crossover corner.**

**Lemma 6.4** (Position of $A$)**.** *Let $m \geq 1$. For $K \neq \Sigma$ the crossover corner $A$ of a level-$m$ $K$-supertile is its vertex 0. For $K = \Sigma$ it lies on side 5 (letter $\delta$), at distance*

$$d_m = 2F_{3m+1}$$

*from vertex 5 and at distance $\delta_m - d_m = 2F_{3m}$ from vertex 0.*

*Proof.* For $K \neq \Sigma$ this is Proposition 2.4(d) for $m \geq 2$, and is read off Fig. 3 for $m = 1$. For $m = 1$, $d_1 = 6$ is read off the level-1 $\Sigma$-supertile. By Proposition 2.4(d) and Table 3, the crossover corner of a level-$(m+1)$ $\Sigma$-supertile follows the children's sides 0.2, 0.1, 7.0 on its side 5. Child 0 is a $\Xi$ and child 7 a $\Gamma$, so these sides have letters $\theta, \epsilon, \alpha$ and $d_{m+1} = \theta_m + \epsilon_m + \alpha_m = (a+2b-2)+(a+b)+(b+2) = 2F_{3m+4}$, in the notation of the previous proof. □

**Lemma 6.5** (Lower bounds)**.** *For every $m \geq 1$: every side has length at least 5; $d_m \geq 6$; and $\delta_m - d_m \geq 4$.*

*Proof.* By Proposition 6.2 all letter lengths increase with $m$, and at $m = 1$ the smallest is $\beta_1 = 5$. Also $d_m \geq d_1 = 6$ and $\delta_m - d_m = 2F_{3m} \geq 2F_3 = 4$. □

## 7. The signature at every level

7.1. **The skeleton as a function of side lengths.** Fix a parent kind $T$. Let $\lambda$ assign a positive integer $\lambda(x)$ to each letter $x$, let $d$ be an integer with $0 < d < \lambda(\delta)$, and let $\omega \in \{\pm 1\}$. Build an abstract composition from Tables 1 and 2. The children are $c = 0, \ldots, n_T - 1$; the rim of child $c$ is a cycle of six sides, side $j$ having length $\lambda$(letter of side $j$ of $K_c$). The child's crossover corner $A_c$ is its vertex 0, or for $K_c = \Sigma$ the point of its side 5 at distance $d$ from vertex 5, and $B_c = A_c + \omega$ in the child's order. Sides are identified in pairs as in Table 2, reversing direction: vertex $j$ of $c$ with vertex $j' + 1$ of $c'$, and the point at distance $o$ from vertex $j$ with the point at distance $\lambda - o$ from vertex $j'$; paired sides have the same letter and hence the same length $\lambda$. The sides marked $\partial$ form the rim, in the order of Table 3. Marks, runs and the skeleton are then defined as in Section 5: junctions are the children's vertices at which the neighbour in Table 2 changes. Call the resulting skeleton $\mathcal{S}_T(\lambda, d, \omega)$.

By Propositions 2.4 and 6.1 and Lemma 6.4, if the children of a level-$(m+1)$ $T$-supertile have the signature at $(A_c, A_c + \omega_m)$ in their own orientation, then the skeleton of its composition (with all children in their own orientation) is $\mathcal{S}_T(\ell_m, d_m, \omega_m)$, where $\ell_m$ also denotes the vector of letter lengths. Indeed: (i) by Proposition 2.4(a),(b) and Proposition 6.1 the children have the kinds and side lengths of the abstract composition, and side $j$ of $X_c$ is side $j'$ of $X_{c'}$ traversed oppositely, so points are matched as above; (ii) the rims of two children meet only in the closed sides they share: at a point inside a side both rim edges are shared with the partner, so the two children fill a neighbourhood, and a vertex lies on exactly three level-$m$ supertiles, any two of which share a side ending there (Theorem 2.1(b)); (iii) so the actual junctions are exactly the children's vertices at which the neighbour in Table 2 changes, each side being wholly shared with one sibling or wholly on the rim; (iv) $A_c$ is placed by Lemma 6.4, and $B_c = A_c + \omega_m$ by hypothesis; (v) the rim of the parent is given by Table 3 (Proposition 2.4(c)). Hence the marks on each child's rim, their order, the runs, partners, lengths and the order of the rim runs are those of $\mathcal{S}_T(\ell_m, d_m, \omega_m)$.

**Lemma 7.1** (Parity lemma)**.** *Say that $(\lambda, d)$ satisfies the* order condition *if $\lambda(x) \geq 3$ for every letter $x$ and $3 \leq d \leq \lambda(\delta) - 3$. If $(\lambda, d)$ and $(\lambda', d')$ both satisfy it, and $\lambda \equiv \lambda'$, $d \equiv d' \pmod 2$, then $\mathcal{S}_T(\lambda, d, \omega) = \mathcal{S}_T(\lambda', d', \omega)$ for every $T$ and $\omega$.*

*Proof.* We use two facts: a vertex of one child that lies on the rim of another is a vertex of that one too (in the abstract composition sides are glued end to end, so this holds by construction), and every composition has exactly one $\Sigma$ child (Table 1). We list the possible marks on a side $s$ of a child, of length $\ell$, by their distance from its first vertex. A junction is a vertex, at distance 0 or $\ell$. A crossover corner that is a vertex 0 is a vertex too, and the corresponding $B$ lies at distance 1 or $\ell - 1$, on side 0 or 5 of its child or on the partner of that side. The crossover corners of the $\Sigma$ child lie at distances $d$ and $d \pm 1$ along its side 5, which has letter $\delta$, or at distances $\ell - d$ and $\ell - d \mp 1$ along the partner of that side. A point inside a side lies only on that side and its partner. So which of these *symbolic positions* carry marks is decided by Tables 1 and 2, by $\omega$, and by which child is the $\Sigma$, and not by $\lambda$ or $d$. Under the

order condition, the symbolic positions that can occur on one side are strictly increasing in the orders

$$0 < 1 < d-1 < d < d+1 < \ell-1 < \ell \qquad \text{and} \qquad 0 < 1 < \ell-d-1 < \ell-d < \ell-d+1 < \ell-1 < \ell,$$

where in both cases $\ell = \lambda(\delta)$ if $d$ occurs. Hence the cyclic order of the marks on each child's rim, and which marks coincide, are the same for $(\lambda, d)$ and $(\lambda', d')$; matching marks with the same symbolic positions is a bijection as in the definition of equal skeletons (Section 5). The partner runs and the order of the rim runs are then determined by Table 2. Finally, the length of a run is a sum of side lengths plus a difference of two symbolic positions, so its parity is determined by $\lambda$ mod 2 and $d$ mod 2. □

7.2. **The induction.**

**Computation 7.2** (Base cases)**.** For $m = 1, 2, 3$ and every kind $K$, the level-$m$ $K$-supertile has the signature at $(A, B)$, where $A$ is as in Lemma 6.4 and $B = A + \omega_m$ in own order. Moreover $\ell_1$ and $d_1$ are as stated in Proposition 6.1 and Lemma 6.4.

This was computed from the tiles, using Lemma 4.2 level by level, in exact arithmetic; the program checks the statement as worded: the two non-hooks are the crossover corners of the $\Gamma$ child, they are adjacent, $B = A + \omega_m$, they carry the crossover, and there is no closed strand. A second, independent program, in floating point, confirms at levels 1 to 9 that every supertile has exactly two adjacent non-hooks, carrying the crossover, which are the crossover corners of its $\Gamma$ child, and no closed strand (Appendix A). The rims have $E_1 = 44, 46$, $E_2 = 160, 182$ and $E_3 = 652, 758$ edges (for $\Gamma$ and the other kinds respectively).

**Theorem 7.3** (Signature Theorem)**.** *In every homochiral tiling by* Tile$(1, 1)$*, every level-$m$ supertile of every kind has the signature at $(A, B)$, where $A$ is its crossover corner (Section 2.3; placed as in Lemma 6.4 for $m \geq 1$) and $B = A + \omega_m$ in its own order. In particular no supertile contains a closed strand.*

*Proof.* For $m = 0$ this is Lemma 4.4, and for $1 \leq m \leq 3$ it is Computation 7.2. Let $m \geq 3$, assume the theorem for all levels up to $m$, and let $Y$ be a level-$(m+1)$ supertile of kind $T$. A level-$(m-1)$ $T$-supertile $Y'$ exists: by Table 1 every supertile of level $n+2$ contains level-$n$ supertiles of all nine kinds (its $\Gamma$ child has children $\Pi, \Delta, \Theta, \Sigma, \Xi, \Phi, \Gamma$, its $\Sigma$ child a $\Lambda$, and its $\Phi$ child a $\Psi$). Its children have the signature, so the skeleton of its composition is $\mathcal{S}_T(\ell_m, d_m, \omega_m)$. By Corollary 6.3 and Lemmas 6.4 and 6.5, $(\ell_m, d_m)$ and $(\ell_{m-2}, d_{m-2})$ satisfy the order condition and are congruent mod 2; also $\omega_m = \omega_{m-2}$. By Lemma 7.1 the skeleton of $Y$ equals $\mathcal{S}_T(\ell_{m-2}, d_{m-2}, \omega_{m-2})$, the skeleton of the composition of $Y'$; let $\iota$ be a bijection of marks as in the definition of equal skeletons. By hypothesis $Y'$ has no closed strand and has hooks at all its rim vertices except its crossover corners, which are those of its $\Gamma$ child $\Gamma'$: $A_{Y'} = A_{\Gamma'}$ by definition, and the rim edge from $A$ towards $B_{Y'}$ lies on side 0 or 5 of $\Gamma'$, which are rim sides of $Y'$ (Table 2), so $B_{Y'} = A + \omega_{m-1}$ in the own order of $Y'$ is $A + \omega_{m-2} = B_{\Gamma'}$ in the opposite own order of $\Gamma'$. By Proposition 5.4 the pairing at the parent marks and the number of closed strands are determined by the skeleton, and $\iota$ carries them from $Y'$ to $Y$, mapping the parent marks of $Y'$ to those of $Y$ in order and $A_{\Gamma'}, B_{\Gamma'}$ to $A_\Gamma, B_\Gamma$; every other rim vertex of $Y$ is a hook. Hence $\kappa(Y) = 0$, and $Y$ has hooks at all its rim vertices except the crossover corners $A_\Gamma, B_\Gamma$ of its $\Gamma$ child, with the crossover $\mathrm{R}(A_\Gamma) \leftrightarrow \mathrm{R}(B_\Gamma)$, $\mathrm{L}(A_\Gamma) \leftrightarrow \mathrm{L}(B_\Gamma)$ there. These two corners are adjacent on the rim of $Y$, since both sides 0 and 5 of the $\Gamma$ child lie on the rim (Table 2). By definition $A_\Gamma$ is the crossover corner of $Y$, placed as in Lemma 6.4. Finally $B_\Gamma = A_\Gamma + \omega_m$ in the child's own order, which is opposite to that of $Y$; so $B = A - \omega_m = A + \omega_{m+1}$ in the own order of $Y$. □

*Remark* 7.4. Since the own orientation alternates with $m$ and so does $\omega_m$, in absolute terms $B$ is always the counter-clockwise neighbour of $A$ on the rim, as it is for the tile ($B = v_1$).

**Computation 7.5** (The induction step directly)**.** As an independent check of the step, the 18 skeletons $\mathcal{S}_T(\ell_1, d_1, \pm 1)$ were built directly from Tables 1 and 2 and the lengths, without

any geometry. In every case the reduced walk of Proposition 5.4 closes no loop and gives hooks at all parent marks except the Γ child's $A$ and $B$, which are adjacent and carry the crossover. Each skeleton has 8 children (7 for Γ), 24 marks (21) and 52 runs (45), of which 10 (9) lie on the rim. The skeletons obtained from $(\ell_2, d_2)$, $(\ell_3, d_3)$ and $(\ell_1 + 2\ell_2, d_1 + 2d_2)$ are identical, as Lemma 7.1 predicts. They also coincide with the skeletons extracted from the actual supertiles at every level from 2 to 9 (this comparison does not distinguish the parent's $A$ from its $B$; the children's are distinguished). Together with Lemma 7.1, this computation alone gives the induction step for every $m \geq 1$, so that only the base case $m = 1$ of Computation 7.2 would be needed; we have preferred the proof above, which rests on the direct computation of levels 2 and 3.

7.3. **No closed strands.** A closed strand is finite. If the supertiles containing it eventually include all its tiles, it is a closed strand of a single supertile, which Theorem 7.3 rules out. In a tiling with a fault, a strand could a priori close up across the fault; the next lemma shows that it would then live where two or three supertiles meet. Write $\beta_m = F_{3m+1} + 2$ for the shortest side length at level $m$ (Proposition 6.2).

**Lemma 7.6** (Localisation Lemma)**.** *Let $C$ be a closed strand of a homochiral tiling, with $r$ pieces, and let $m \geq 1$ with $\beta_m > r$. Then the tiles of $C$ lie in a single level-$m$ supertile, or in the union of two level-$m$ supertiles that share a side, or in the union of three level-$m$ supertiles around a common vertex.*

*Proof.* Follow $C$ and cut it at the tile edges where it passes from one level-$m$ supertile to another. If there are none, we are in the first case. Otherwise $C$ is a cyclic sequence of arcs of level-$m$ supertiles $Y$ (Lemma 4.1), joined across *crossing edges*, each of which lies on the rims of two different supertiles. Each arc is a hook or a crossover arc of $Y$ (Theorem 7.3), so the two crossing edges at its ends are rim edges of $Y$ with a common vertex: for a hook at $u$ they are the two rim edges at $u$, and for a crossover arc they are two of the three rim edges at $A$ and $B = A \pm 1$. Hence the crossing edges form a connected graph $G$ with at most $r$ edges. It lies in the *skeleton*, the union of the rims of all level-$m$ supertiles, which by Theorem 2.1(b) is a graph in which the vertices of the supertiles have degree 3, all other points degree 2, and any path between two different supertile vertices runs along a whole side, of at least $\beta_m > r$ edges. So $G$ contains at most one supertile vertex $u$, and lies in one side, or in the union of the three sides at $u$. Every level-$m$ supertile containing a tile of $C$ is entered by $C$ across a crossing edge on its rim. If $G$ lies in one side, the only such supertiles are the two that share it. Otherwise they are among the three supertiles around $u$, each side at $u$ being shared by two of them. □

In the second and third cases, $C$ is a closed strand of the union $U$ of two or three level-$m$ supertiles, each with the signature at its crossover corners, placed as in Lemma 6.4 with $B = A + \omega_m$ (Theorem 7.3); their shared sides carry matching labels (Theorem 2.1(b)), so have equal lengths. As for $\mathcal{S}_T(\lambda, d, \omega)$ in Section 7, the skeleton of $U$, viewed as a composition of these two or three children, is determined by their kinds, the sides glued, $\ell_m$, $d_m$ and $\omega_m$, and by Proposition 5.4 so is the number of closed strands of $U$.

**Computation 7.7** (Where supertiles meet)**.** Consider every pair of kinds and sides $(K, j)$, $(K', j')$ whose labels match (the same letter, with opposite signs, or both $\eta$), giving two supertiles glued along side $j$ of the first and $j'$ of the second (264 cases), and every triple $(K_i, j_i)_{i=1,2,3}$ giving three supertiles around a common point that is vertex $j_i$ of the $i$th, with side $j_i$ of each glued to side $j_{i+1} - 1$ of the next, all with matching labels (489 cases). For every level $m \geq 1$, the union has no closed strand, except for the three Γ-supertiles around a point that is vertex 5 of each when $m$ is odd, or vertex 1 of each when $m$ is even; that union has one closed strand.

This was computed symbolically, with the letter lengths $\ell_m$ as variables and $d_m = \zeta_m$ (both are $2F_{3m+1}$, by Proposition 6.2 and Lemma 6.4), once for each parity of $m$, recording every

comparison of lengths as an obligation and certifying the obligations for every $m \geq 1$ of that parity as in the second implementation of Lemma 8.8; for $1 \leq m \leq 14$ the computation with the actual lengths takes the same branches. Applied to the nine full compositions, the same program finds no closed strand, as Theorem 7.3 requires, and it does find closed strands in unions glued against the signs of the matching rule (Appendix A).

**Lemma 7.8** (Three Gammas Lemma)**.** *In no homochiral tiling do three level-$m$ $\Gamma$-supertiles ($m \geq 1$) meet at a point that is vertex 5 of each when $m$ is odd, or vertex 1 of each when $m$ is even.*

*Proof.* At every level a $\Gamma$-supertile occurs only as the last child of its parent (Table 1), so three of them have three different parents. The common point lies on the rims of all three parents, so it is a vertex of each (Theorem 2.1(b)). But by Table 3 vertex 5 of the last child is the point between the pieces at positions 0 and 1 of the parent's rim, and vertex 1 is the point before the piece at position 21 (position 19 for a $\Gamma$ parent); none of these positions is the first position of a side, so the point lies inside a side of the parent and is not a vertex. □

*Proof of the Endless Strand Theorem.* Let $C$ be a closed strand of a homochiral tiling, with $r$ pieces, and choose $m \geq 1$ with $\beta_m > r$. By Lemma 7.6, $C$ is a closed strand of one level-$m$ supertile, which Theorem 7.3 excludes, or of a union of two or three level-$m$ supertiles as in Computation 7.7. That union is not the exceptional one, by the Three Gammas Lemma (Lemma 7.8), so it has no closed strand: a contradiction. By Lemma 3.4, every strand is a two-way infinite path. □

## 8. The nest tiling and the single strand

8.1. **The nest.** Child 1 of every supertile is a $\Delta$ (Table 1), and child 1 of a $\Delta$ is again a $\Delta$.

**Construction 8.1** (The nest)**.** Let $\mathcal{T}$ be a homochiral tiling [17, Thm. 2.2]. For each $h \geq 0$, let $Y^{(h)}$ be a level-$h$ $\Delta$-supertile of $\mathcal{T}$; one exists, for instance child 1 of any level-$(h+1)$ supertile. Let $Y^{(h)} = Y^{(h)}_h \supset Y^{(h)}_{h-1} \supset \cdots \supset Y^{(h)}_0$ be its chain of descendants, each child 1 of the next. Let $g_h$ be the proper motion taking the tile $Y^{(h)}_0$ onto Tile(1, 1) (with $v_0$ at the origin), and put $N^{(h)}_b = g_h(Y^{(h)}_b)$ for $b \leq h$.

**Proposition 8.2.** *$N^{(h)}_b$ does not depend on $h \geq b$ (or on the choices made). Writing $N_b$ for it, $N_0 = \{\text{Tile}(1,1)\}$, and $N_b$ is child 1 of $N_{b+1}$ for every $b$.*

*Proof.* Let $h, h' \geq b$. By Lemma 2.6 there is a proper motion $f$ with $f(Y^{(h)}_b) = Y^{(h')}_b$ that maps descendants to descendants with their numbers; hence $f(Y^{(h)}_{b'}) = Y^{(h')}_{b'}$ for all $b' \leq b$, in particular for the tiles $Y_0$. So $g_{h'}f$ and $g_h$ both take the tile $Y^{(h)}_0$ onto Tile(1, 1), and since a tile has no symmetry they are equal. Thus $N^{(h')}_b = g_{h'}f(Y^{(h)}_b) = N^{(h)}_b$. □

The *nest tiling* is $\mathcal{N} = \bigcup_{b \geq 0} N_b$, a set of tiles. Figure 10 shows three consecutive blocks.

**Lemma 8.3** (Grandparent Lemma)**.** *For every $b \geq 1$, the block $N_b$ does not meet the rim of $N_{b+2}$.*

*Proof.* By Table 2, the sides 0, 1, 3, 4, 5 of child 1 face the siblings 7, 0, 2, 3, 4, and side 2 lies on the parent's rim. A point of $N_b$ that is not on its closed side 2 is therefore an interior point of $N_{b+1}$. This is clear for interior points of $N_b$ and for interior points of the other sides, where $N_b$ and one sibling together fill a neighbourhood. At the vertices 0, 1, 4, 5 of $N_b$, exactly three level-$b$ supertiles meet; two of them are the distinct siblings across the adjacent sides, and together they fill a neighbourhood. Hence

$$N_b \cap \partial N_{b+1} = \text{closed side 2 of } N_b. \tag{2}$$

By Table 3 (the $\Delta$ row: positions 14 to 18), side 4 of the parent is the concatenation of 2.2, 2.1, 1.2, 0.4, 0.3. Since 1.2 is neither first nor last, the closed side 2 of $N_b$ lies in the relative interior of side 4 of $N_{b+1}$. Now let $p \in N_b \cap \partial N_{b+2}$. Then $p \in \partial N_{b+1}$, for otherwise

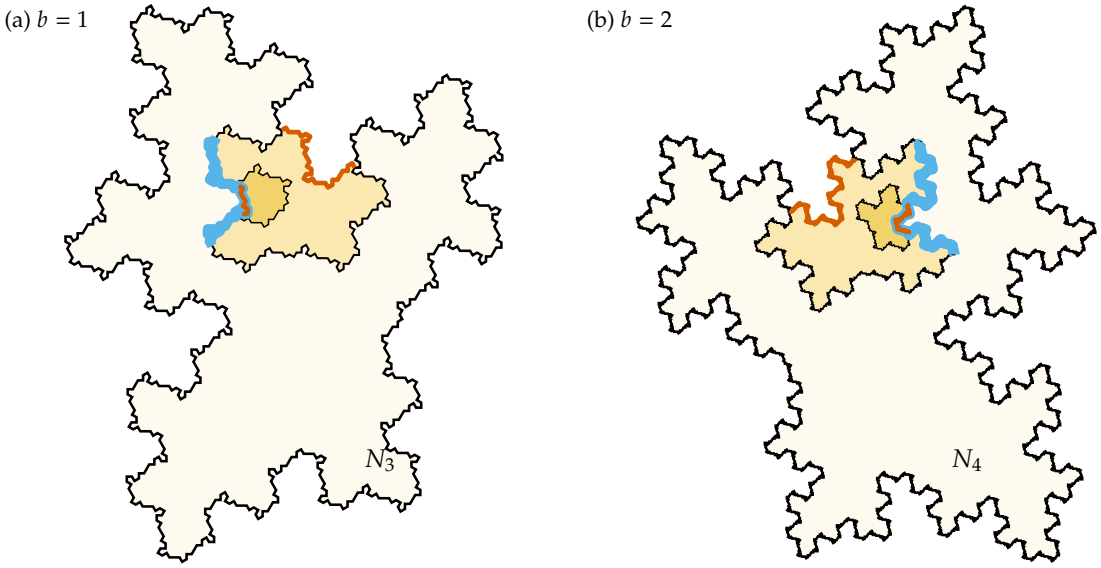



FIGURE 10. Three consecutive blocks of the nest: (a) $N_1 \subset N_2 \subset N_3$ and (b) $N_2 \subset N_3 \subset N_4$. Vermilion: side 2 of the two smaller blocks, which is the only side of each block on the rim of the next. Blue: side 4 of the middle block. Since the vermilion side of the smallest block lies inside the blue side, and the blue side faces a sibling inside the largest block, the smallest block does not reach the rim of the largest (Lemma 8.3).

$p$ would be interior to $N_{b+1} \subset N_{b+2}$. So $p$ lies on side 4 of $N_{b+1}$ by the above, and, by (2) for $N_{b+1}$, on the closed side 2 of $N_{b+1}$. But the closed sides 2 and 4 of $N_{b+1}$ are disjoint: they are arcs of a simple closed curve separated by side 3 on one side and by sides $5, 0, 1$ on the other. □

**Proposition 8.4.** *$\mathcal{N}$ is a homochiral tiling of the plane, and every tile of $\mathcal{N}$ lies in some $N_b$.*

*Proof.* Any two tiles of $\mathcal{N}$ lie in a common block, so their interiors are disjoint. We show that the tiles cover the plane. Fix $h$ and work in the homochiral tiling $\mathcal{T}_h = g_h(\mathcal{T})$, which contains $N_0, \dots, N_h$. Call two tiles of $\mathcal{T}_h$ *neighbours* if they share a point, and let dist be the resulting graph distance. We claim that every tile at distance at most $k$ from $N_1$ lies in $N_{1+2k}$, provided $1 + 2k \le h$. For $k = 0$ there is nothing to prove. If $t$ has distance $k + 1$, it shares a point $p$ with a tile of $N_{1+2k}$. By the Grandparent Lemma (Lemma 8.3), $p$ is an interior point of $N_{3+2k}$, so the interior of $t$ meets the interior of a tile of $N_{3+2k}$; hence $t$ is that tile. Now let $x \in \mathbb{R}^2$ and let $D$ be the disc of radius $r = |x| + 1$ about the origin. The tiles of $\mathcal{T}_h$ meeting $D$ have diameter $3 + \sqrt{3} < 5$ and area $3 + 3\sqrt{3} > 8$, so there are at most $k_0 = \lfloor \pi(r + 5)^2/8 \rfloor$ of them. They cover the connected set $D$, so their neighbour graph is connected. One of them is $N_0 \subset N_1$, since the origin is a vertex of $N_0$. So every tile of $\mathcal{T}_h$ meeting $D$, in particular one containing $x$, lies in $N_{1+2k_0}$ as soon as $h \ge 1 + 2k_0$. Since $N_{1+2k_0}$ does not depend on $h$, $x$ is covered by $\mathcal{N}$. □

The blocks $N_b$ are supertiles in the sense of Section 2: each is congruent, with all its structure, to a level-$b$ $\Delta$-supertile of $\mathcal{T}$. So Theorem 7.3 applies to them. In particular $\mathcal{N}$ has no closed strand: a closed strand passes through finitely many tiles, which lie in a common block $N_b$ by Proposition 8.4, and would be a closed strand of $N_b$.

8.2. **Merging.** For $b \ge 1$ and $k \ge 0$, every arc of $N_b$ is contained in a unique arc of $N_{b+k}$, because $N_{b+k}$ has no closed strand. Let $\mathcal{P}_k(b)$ be the partition of the $E_b(\Delta)$ arcs of $N_b$ in

| $b$ | tiles in $N_b$ | arcs of $N_b$ | $k=1$ | $k=2$ | $k=3$ | $k=4$ |
|---|---|---|---|---|---|---|
| 0 | 1 | 14 | 5 | 2 | 1 | 1 |
| 1 | 9 | 46 | 10 | 2 | 2 | 1 |
| 2 | 71 | 182 | 37 | 2 | 1 | 1 |
| 3 | 559 | 758 | 146 | 2 | 2 | 1 |
| 4 | 4,401 | 3,198 | 613 | 2 | 1 | 1 |
| 5 | 34,649 | 13,534 | 2,586 | 2 | 2 | 1 |
| 6 | 272,791 | 57,318 | 10,949 | 2 | 1 | – |
| even $b \geq 2$ | | | | 2 | 1 | 1 |
| odd $b \geq 3$ | | | | 2 | 2 | 1 |

TABLE 5. The number of arcs of $N_{b+k}$ on which the arcs of $N_b$ lie, that is, the number of blocks of $\mathcal{P}_k(b)$ (row $b = 0$: the 14 pieces of the single tile, computed on the drawing up to level 5). Rows $1 \leq b \leq 6$: computed by following every ribbon, up to level 9 (in floating point; for $b \leq 3$ also in exact arithmetic, up to level 7). Last two rows: Theorem 8.5 and its proof, for every $b$.

which two arcs are in the same block when they lie on the same arc of $N_{b+k}$. The blocks only merge as $k$ increases.

**Theorem 8.5** (Merging)**.** *For every $b \geq 1$, $\mathcal{P}_4(b)$ has a single block; if $b$ is even, so does $\mathcal{P}_3(b)$.*

Table 5 shows the numbers. The proof follows the arcs of $N_b$ through the levels using Proposition 5.4 and Corollary 5.5, and shows that this computation is the same for all $b$ of the same parity.

*Label states.* Fix $b \geq 1$. We follow the arcs of $N_b$ up the nest. By Theorem 7.3 every arc of $N_{b+k}$ is a hook at a rim vertex other than the crossover corners, or one of the two crossover arcs.

**Definition 8.6** (Label state)**.** Let $k \geq 0$ and $N = N_{b+k}$. A *label state* at relative level $k$ is a quadruple $(H, S, X, \sim)$, where

- $H$ is a finite set of pairs $(u, \lambda)$, with $u$ a rim vertex of $N$ other than $A$ and $B$, the $u$ distinct, and $\lambda$ a *label*;
- $S$ is a set of pairwise disjoint *stretches*: arcs of consecutive rim vertices of $N$ containing neither $A$, nor $B$, nor any $u$ with $(u, \lambda) \in H$;
- $X = (x_{\mathrm{R}}, x_{\mathrm{L}})$, each entry a label or empty;
- $\sim$ is an equivalence relation on the labels occurring in $H$ and $X$.

Each label $\lambda$ stands for a set $a(\lambda)$ of arcs of $N_b$, the sets of distinct labels being disjoint. The state *represents* $\mathcal{P}_k(b)$ if (i) for every vertex of a stretch, the hook of $N$ there contains exactly one arc of $N_b$; (ii) for $(u, \lambda) \in H$, the hook of $N$ at $u$ contains exactly the arcs in $a(\mu)$ for $\mu \sim \lambda$; (iii) the crossover arc of $N$ through $\mathrm{R}(A)$ (through $\mathrm{L}(A)$) contains exactly the arcs in $a(\mu)$ for $\mu \sim x_{\mathrm{R}}$ (for $\mu \sim x_{\mathrm{L}}$), and no arc of $N_b$ if the entry is empty; (iv) every arc of $N_b$ lies in exactly one of the sets of (i)–(iii).

If a state represents $\mathcal{P}_k(b)$, the blocks of $\mathcal{P}_k(b)$ are the $\sim$-classes (one for each labelled arc of $N$) and the singletons of the stretch vertices, so

$$|\mathcal{P}_k(b)| = (\text{number of } \sim\text{-classes}) + (\text{number of stretch vertices}). \tag{3}$$

At $k = 0$ the state with $H = \varnothing$, one stretch consisting of all rim vertices of $N_b$ other than $A$ and $B$, two labels $x_{\mathrm{R}}, x_{\mathrm{L}}$ standing for the two crossover arcs, and $\sim$ trivial, represents $\mathcal{P}_0(b)$.

*The step.* Let a state at level $k$ be given, and pass to $N^+ = N_{b+k+1}$, in which $N = N_{b+k}$ is child 1; the skeleton of the composition is $\mathcal{S}_\Delta(\ell_{b+k}, d_{b+k}, \omega_{b+k})$ (Section 7). Positions on the rim of $N$ are its positions as child 1.

(S1) For $(u, \lambda) \in H$: if $u$ is a mark of child 1, put $\lambda$ at the reduced port $\mathrm{R}_1(u)$, which lies on the hook of child 1 at $u$; if $u$ lies inside a rim run of child 1, put $(u, \lambda)$ into $H^+$; if $u$ lies inside an interior run, put $\lambda$ at the reduced port named in Corollary 5.5.
(S2) Split each stretch at the marks of child 1 that it contains. Each such mark $u$ gets a fresh label, standing for the arc of $N_b$ in the hook at $u$, which is put at $\mathrm{R}_1(u)$. Each maximal part between these marks lies inside one run. If it is a rim run, the part is put into $S^+$. If it is an interior run, then by Corollary 5.5 the hooks of the part lie on at most two arcs of $N^+$, one for each parity of their offsets in the run; for each parity that occurs, a fresh label, standing for the arcs of $N_b$ in those hooks, is put at the reduced port named there. Which parities occur depends only on whether the part has one vertex or more, and on the parity of its first offset.
(S3) Put $x_{\mathrm{R}}$ at $\mathrm{R}_1(A_1)$ and $x_{\mathrm{L}}$ at $\mathrm{L}_1(A_1)$ (if not empty).
(S4) Follow the reduced walk of Proposition 5.4 from every reduced port that carries a label. Labels that reach the same arc of $N^+$ become equivalent. Put into $H^+$ the pair $(u, \lambda)$ for each parent mark $u \neq A^+, B^+$ whose hook in $N^+$ carries a label $\lambda$, and let $X^+$ be the labels on the crossover arcs of $N^+$ through $\mathrm{R}(A^+)$ and $\mathrm{L}(A^+)$.

**Lemma 8.7.** *If the state at level $k$ represents $\mathcal{P}_k(b)$, the state that the step produces represents $\mathcal{P}_{k+1}(b)$.*

*Proof.* By Proposition 5.4 and Corollary 5.5, each hook and crossover arc of child 1 lies on the arc of $N^+$ named in (S1)–(S4); the walk ends at a parent port, since $N^+$ has no closed strand (Theorem 7.3). A hook of child 1 at a vertex inside a rim run is a hook of $N^+$ at the same vertex (Proposition 5.4(a)), so carried pairs and carried stretch vertices keep properties (ii) and (i). By Theorem 7.3 every arc of $N^+$ is a hook or a crossover arc, so every labelled arc of $N^+$ is recorded in $H^+$ or $X^+$; a parent mark is not inside a rim run of child 1, so the pairs added in (S4) are at vertices different from the carried ones, and from the carried stretches. Every element of the old state is placed exactly once, so (iv) persists. □

**Lemma 8.8** (Uniformity in $b$)**.** *For $b \geq 2$ and $1 \leq k \leq 8$, the label state at relative level $k$, with every position written as an integer affine function of $x = \ell_{b-1}$, and the partition $\sim$, depend on $b$ only through the parity of $b$.*

*Proof.* Let $x \in \mathbb{Z}^{54}$, indexed by kind and side, be the vector of side lengths one level below the base. By Proposition 2.4(c), $\ell_{m+1} = M\ell_m$ for $m \geq 1$, where $(M\lambda)(T, J) = \sum \lambda(K_c, j)$ over the pieces $c.j$ of side $J$ of $T$ in Table 3; and $d_{m+1} = \ell_m(\Xi, 2) + \ell_m(\Xi, 1) + \ell_m(\Gamma, 0)$ by the proof of Lemma 6.4. So $\ell_{b+k}$ and $d_{b+k}$ are integer linear functions of $x$, and every position used in the steps is an integer affine function of $x$: the rim positions of the vertices and marks of the children in their own order, the offsets of labelled hooks and stretch endpoints inside runs, and the positions in $N^+$ of carried points. The skeletons depend only on the parity of $b$, by Lemmas 6.5 and 7.1, and parities of positions are fixed by Corollary 6.3 and Lemma 6.4. Besides these, the steps make the following decisions that depend on the lengths: (1) on which side of a child a position lies; (2) whether a crossover corner of a child is a vertex of that child; (3) the bounds $d_{b+k} \geq 2$ and $\ell_{b+k}(\Sigma, 5) - d_{b+k} \geq 2$; (4) whether two candidate mark positions coincide; (5) the strict cyclic order of the marks on each child's rim, and of the marks of child 1 inside a stretch; (6) whether a point is a mark of child 1, or lies strictly inside a given run, and at which offset; (7) whether a mark lies in a stretch; (8) whether a part of a stretch is empty, and whether it is a single vertex; (9) whether a parent mark is the parent's corner $A^+$; (10) whether the positions of labelled hooks are distinct; (11) the reduction of a position modulo a rim length. Each is an inequality $f(x) \geq t$ or an equality $f(x) = 0$ for an integer affine function $f$; a disequality is recorded as $\pm f(x) \geq 1$, and a reduction modulo a rim length $E$, with a fixed quotient $q$, as $0 \leq u - qE$ and $u - qE < E$. (Three further comparisons in our program are not recorded, because their outcome is the same for every $b$: whether $B = A + \omega$ must be reduced modulo the rim, which for children other than $\Sigma$ depends only on the sign of $\omega$ and for $\Sigma$ never happens, by the recorded bound

$d \geq 2$; and two tests in the map from a child's rim positions to the parent's, which for the non-negative positions passed to it are decided by the recorded side obligations.) At each decision the procedure for a given $b$ evaluates a predicate that is determined by the outcomes of the earlier decisions. So if every recorded inequality and equality holds for $b$, then, by induction on the number of decisions, the procedure for $b$ takes the same branches as the symbolic run below and produces the same label states.

We ran the computation symbolically, carrying every position as an integer affine function of $x$, once for each parity of $b$. At each decision the branch was chosen by evaluating at $x = \ell_1$ (the side lengths of $b = 2$), with $\omega_{b+k}$ set by the parity being run; for odd $b$ this point is a stand-in, not the data of an actual base, and the certificates below show that $b = 3$ takes the same branches. Over 8 relative levels this records 1,068 obligations for even $b$ and 1,038 for odd $b$ (2,106 in all, 1,407 distinct); 4 of the 2,106 (3 of the distinct ones) are equalities. To certify one, write $f(x) = a \cdot x + e$. Of the obligations, 1,001 + 979 either vanish identically (2 equalities) or have $a \geq 0$ and $5 \sum a + e \geq t$, so they hold for every $b \geq 2$ by Lemma 6.5. For the remaining 67 + 59 = 126 (118 distinct, 2 of them equalities), we use Proposition 6.2: in letter coordinates

$$\ell_m = F_{3m+1}\, u + F_{3m+2}\, w + z,$$
$$u = (0,1,0,0,1,2,0,1), \quad w = (1,0,2,2,1,0,2,2), \quad z = (2,2,0,0,0,0,0,-2),$$

in the order $\alpha, \dots, \theta$. So each obligation reads $g(n) = PF_n + QF_{n+1} + R \geq 0$ for all $n = 3b-2 \geq 4$, with integers $P, Q, R$. By Binet's formula $PF_n + QF_{n+1} = \big((P+Q\varphi)\varphi^n - (P+Q\psi)\psi^n\big)/\sqrt{5}$, with $\varphi = (1+\sqrt{5})/2$ and $\psi = -1/\varphi$. If $P = Q = 0$ the obligation is $R \geq 0$. Otherwise $P + Q\varphi$ and $P + Q\psi$ are non-zero, since $\varphi$ is irrational; if $P + Q\varphi < 0$ the obligation fails for large $n$, and we check that this does not happen. Since $|P+Q\varphi|\,|P+Q\psi| = |P^2+PQ-Q^2| \geq 1$, we get $g(n) \geq \varphi^n/(\sqrt{5}\,c) - c/\sqrt{5} - |R|$ with $c = |P| + |Q|$. This is non-negative once $\varphi^n \geq c(c + 3|R|)$ (the program tests $F_{n-1} \geq c(c + 3|R|)$, which implies it), and the finitely many smaller $n$ are checked directly. Equalities require $P = Q = R = 0$. All 126 obligations pass. As a second, independent check, each was also certified on the space $\operatorname{span}\{M^i \ell_1\}$, which has dimension 3 and contains every $x = \ell_{b-1}$: an inequality by exhibiting $a' \geq 0$ with $a' - a$ orthogonal to that space and $5 \sum a' + e \geq t$, an equality by checking that $a$ is orthogonal to it and $e = 0$, in exact rational arithmetic. So for every $b \geq 2$ every recorded obligation holds, and by the first paragraph the label states are those of the symbolic run of the same parity.

A second implementation, written separately in Python from the definitions and tables of this paper and sharing no code with the first, repeats the symbolic run with the eight letter lengths one level below the base as variables, taking its branches at the actual bases $b = 2$ and $b = 3$. It records 150 distinct obligations for each parity and certifies them by a third argument: an obligation becomes $h_j + R \geq 0$ with $h_j = PF_{n_j} + QF_{n_j+1}$, $n_j = n_0 + 6j$, and $h_{j+2} = 18h_{j+1} - h_j$ (since $F_{n+12} = 18F_{n+6} - F_n$), so once $0 \leq h_j \leq h_{j+1}$ the sequence $h$ never decreases. It reaches the same numbers of classes, checks at every step that the labels and stretches account for every arc of $N_b$ exactly once, and rejects deliberately corrupted inputs. □

*Proof of Theorem 8.5.* For $b = 1$ the partitions were computed directly, by following every ribbon in exact arithmetic (Appendix A): $\mathcal{P}_4(1)$ has a single block. For $b \geq 2$, by Lemma 8.7 and induction on $k$ the label states represent $\mathcal{P}_k(b)$, and by Lemma 8.8 they are those of the symbolic run of the same parity. In those runs the number of $\sim$-classes at relative levels $1, 2, 3, 4$ is $4, 2, 1, 1$ for even $b$ and $3, 2, 2, 1$ for odd $b$, and there is no stretch from relative level 2 on. By (3), $\mathcal{P}_3(b)$ (even $b$) and $\mathcal{P}_4(b)$ (odd $b$) have a single block. (At $k = 1$ a stretch is still carried, and its vertices are singleton blocks; this is why the numbers of Table 5 are larger there, for instance $37 = 4 + 33$ for $b = 2$.) The direct computations for $b = 2$ and $b = 3$ give the same numbers. Once $\mathcal{P}_k(b)$ has a single block, so does $\mathcal{P}_{k'}(b)$ for every $k' \geq k$. □

*Remark* 8.9. The merging was also computed on the actual blocks for $b \le 6$, following every ribbon up to level 9; the results are Table 5. For $b \le 3$ a second, independent program in exact arithmetic follows every ribbon up to level 7, which covers $k \le 4$. The label input of the symbolic run agrees with the one rebuilt from the pattern, the parities and $\omega$, and that in turn agrees with the label input of the actual blocks, $b \le 6$, at every relative level up to level 9.

8.3. **One strand.**

*Proof of the Ariadne Theorem.* By Lemma 3.4, every component of the strand graph $\mathcal{G}(\mathcal{N})$ is a cycle or a two-way infinite path, and there are no cycles, as shown after Proposition 8.4. It remains to show that $\mathcal{G}(\mathcal{N})$ is connected. Let $p, q$ be pieces of tiles $t, t'$ of $\mathcal{N}$. By Proposition 8.4 both tiles lie in some $N_b$ with $b \ge 1$. Since $N_b$ has no closed strand, $p$ and $q$ lie on arcs of $N_b$. By Theorem 8.5 these arcs lie on one arc of $N_{b+4}$, so $p$ and $q$ are on the same strand. The strand passes through every tile, since every tile has pieces. □

## 9. Scope and open questions

**Other tilings.** The Endless Strand Theorem holds for every homochiral tiling. The Ariadne Theorem is proved for the nest tiling, whose chain of supertiles is $\Delta \subset \Delta \subset \cdots$ with child number 1 at every step. Every tiling whose hierarchy covers the plane has some such chain $(K_m, c_m)_{m \ge 1}$, where $X_m$ is a level-$m$ supertile of kind $K_m$ and child $c_m$ of $X_{m+1}$. The induction of Section 7 does not see the chain, but the merging of Section 8 does: its finite part depends on the sequence $(K_m, c_m)$, and so may its outcome.

*Remark* 9.1 (Three strands: a computation, not a theorem). Let $K_m = \Theta$ for odd $m$ and $K_m = \Gamma$ for even $m$, with $c_m = 2$ for odd $m$ and $c_m = 7$ for even $m$ (Table 1). A tiling with this chain is constructed as in Construction 8.1, and its hierarchy covers the plane: the argument of Lemma 8.3 and Proposition 8.4, applied with Tables 2 and 3, shows that no $X_m$ meets the rim of $X_{m+2}$. In this tiling, following the ribbons through the actual supertiles in exact arithmetic, the arcs of $X_1$ lie on exactly three arcs of $X_m$ for $3 \le m \le 8$, and the same holds for the arcs of $X_2$ and of $X_3$ for $5 \le m \le 8$. The label bookkeeping of Section 8, run along this chain, agrees and still gives three at $m = 20$. If this persists for every block and every level, the pieces of this tiling form exactly three strands, each a two-way infinite path by the Endless Strand Theorem. We have not proved this, and the paper does not use it. For the other alignment ($K_m = \Gamma$, $c_m = 7$ for odd $m$ and $K_m = \Theta$, $c_m = 2$ for even $m$) the arcs of $X_1$ lie on a single arc of $X_4$. Along 90,000 chains chosen at random step by step (30,000 for each $b \le 3$), the label bookkeeping put the arcs of $X_b$ on a single arc of $X_m$ for some $m \le 20$ in all but three. The programs are in `experiments/`, with logs; the label bookkeeping along a chain is `hierarchy/chains_label.js`, which agrees with exact strand following on 60 chains up to level 8.

(1) Which homochiral tilings whose hierarchy covers the plane carry a single strand? Is the number of strands always finite, and which numbers occur?

**Faults.** Some homochiral tilings contain two or more infinite supertiles. This follows by the usual compactness argument; we sketch it. For each $m \ge 1$, apply a proper motion to a homochiral tiling so that every tile is $\rho^k \mathrm{Tile}(1,1) + t$ ($k \in \mathbb{Z}$, $t \in \mathbb{Z}[\rho]$) and the origin is a vertex of a level-$m$ supertile; call the result $\mathcal{T}_m$. The origin is then where three level-$m$ supertiles meet [17, Thm. 5.1], and it lies on the boundary of a level-$n$ supertile for every $n \le m$. For fixed $r$ and $n$, since the tiles meet edge to edge, the origin is a tile vertex, and boundedly many tiles meet the disc of radius $r$ about the origin, there are only finitely many possibilities for these tiles, together with their grouping into level-$n'$ supertiles for $n' \le n$, the kinds of these supertiles and their child numbers. By a diagonal argument, some subsequence of the $\mathcal{T}_m$ has all these data eventually constant, for every $r$ and $n$. The limit is a homochiral tiling whose level-$n'$ groupings are compositions as in Proposition 2.4; by the

uniqueness in Theorem 2.1(a) they are its supertiles. In it the origin lies on the boundary of a level-$n$ supertile for every $n$. The finitely many tiles at the origin cover a neighbourhood of it, so they never all lie in one supertile, and they meet at least two different infinite supertiles. The Endless Strand Theorem covers these tilings too: no strand closes up, across a fault or anywhere else. The one configuration in which a closed strand could form where supertiles meet, three Γ-supertiles around a common vertex (Computation 7.7), never occurs (Lemma 7.8). The Ariadne Theorem says nothing about tilings with faults.

**Other drawings and designs.** The theorems concern only the pairing of ports (Definition 3.1) and the straight-through rule. They therefore apply to any drawing that realises these, including drawings on the curved Spectre, whose tilings correspond to the homochiral tilings by Tile(1, 1) [17, §2]. The method uses the decoration only through the signature of the tile (Lemma 4.4), the position of the crossover, which fixes $A$ (Lemma 6.4), the base cases (Computation 7.2) and, for the Ariadne Theorem, the computations of Section 8. For any other decoration in which the tile has the signature at some edge, the same finite computations can be run; their outcome may differ.

## Appendix A. The computations

All programs and their logs are published with this paper at `https://cortado-j.github.io/SPECTRE_KNOT/`, folder `code/`. One command, `./run_all.sh`, re-runs everything the proof needs in about twelve minutes on a four-core machine; `./run_all.sh --all` adds the corroborating computations to level 9, and takes about an hour in all (with the memory of the JavaScript programs capped at 6.5 GB). The file `code/README.md` maps every statement of this paper to the script that checks it.

**Two implementations.** The proof uses two kinds of computation. The skeleton walks, the symbolic run and the certificates work with integers and rationals only; the integers involved are below $2 \cdot 10^7$ in absolute value, far below $2^{53}$, so the parts written in JavaScript are exact as well. The geometric computations on actual supertiles that the proof uses are the Mystic (Lemma 4.4), the level-1 data (vertex positions, $\ell_1$, $d_1$, the position of $A$), the patterns of the compositions $m \to m+1$ for $m \le 3$, the base cases (Computation 7.2) and the merging for $b = 1$ (Theorem 8.5). All of them are done in exact arithmetic by `exact/exact.py`, in Python. It represents every vertex exactly as an element $a + b\rho + c\rho^2 + d\rho^3$ of $\mathbb{Z}[\rho]$, $\rho = e^{i\pi/6}$, with integers $a, b, c, d$, and every motion as $z \mapsto \rho^k z + t$ (or $\rho^k \bar{z} + t$ in reflected frames). Floating point is used only to read off the sign of the area of a polygon whose area exceeds 8 in absolute value. The inputs of the symbolic run (the pattern, the matrix $M$ and $\ell_1$) are read from the JavaScript programs and checked equal to the exact ones. The symbolic run is also done a second time, by `second/nest2.py`, in Python, which takes the pattern from the tables of this paper and shares no code with the first (see the end of the proof of Lemma 8.8); all the programs were written by the same AI system.

The JavaScript programs in `core/` and most of `hierarchy/` repeat these computations independently and extend them: the patterns for $m \le 7$, the signature up to level 9, the merging for $b \le 6$ up to level 9. These extensions corroborate the proof but are not used in it. The programs share no code with `exact.py`, although both implement the key-point construction of [17, App. A] and so the same table of placements. They work in double-precision floating point and identify points by rounding coordinates to $10^{-2}$ or $10^{-3}$. We do not bound their rounding error. A separate check (`referee/overlap.js`), which builds the supertiles of levels 1 to 7 in the same way, finds distinct vertices at distance at least 0.89, and points that it identifies within $4 \times 10^{-12}$ of each other; and up to level 9 every rim found has exactly $E_m$ edges (Proposition 6.2) and every rim vertex has degree 2, which a wrongly identified point would break. All programs compose the ribbon pairings with Lemma 4.2 and nothing else. (In the programs the sign $\omega_m$ is called `beta`.)

**Sizes.** The script `hierarchy/sizes.py` computes these numbers from Table 1 and Proposition 6.2. They agree with the supertiles built by the programs (tile counts up to level 7, rim lengths up to level 9).

| level $m$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| tiles in a $\Delta$ | 9 | 71 | 559 | 4401 | 34649 | 272791 | 2147679 | 16908641 | 133121449 |
| rim $E_m(\Delta)$ | 46 | 182 | 758 | 3198 | 13534 | 57318 | 242790 | 1028462 | 4356622 |
| shortest side $\beta_m$ | 5 | 15 | 57 | 235 | 989 | 4183 | 17713 | 75027 | 317813 |

**What was computed.**

| statement | computation | script |
|---|---|---|
| Definition 3.1 and Lemma 4.4 | the tile's pairing; the Mystic | `core/tile.js`, `exact/exact.py` |
| Proposition 2.4, Remark 2.5 | vertices of levels 1–6 (points where three supertiles meet); the pattern of compositions $m \to m+1$, $m = 1, \ldots, 7$, all identical; exact for $m \leq 3$ | `hierarchy/vertices.js`, `combin.js`, `exact/exact.py` |
| Proposition 2.4 (base of $(*)$) | the level-1 supertiles as hat/turtle clusters, fitted to [17, Fig. 4.1] (dots, outline, shading; labels read by hand) | `referee/fig41_compare.py`, `fig41_render.py` |
| Remark 2.5 | agreement with an encoding of [17, Fig. 5.1], all nine kinds; its dots re-found from the PDF; two corrupted encodings rejected | `referee/fig51_compare.py`, `fig51_raster.py` |
| Remark 2.5 | tiles of every supertile, levels 1–7, have disjoint interiors and fill its rim | `referee/overlap.js` |
| Propositions 6.1 and 6.2, Corollary 6.3, and Lemma 6.4 | letters, recurrence, lengths at levels 1–8, closed forms | `hierarchy/recur.js`, `letters.py` |
| Computation 7.2 | signature at levels 1–3 (and 4): exact, and in floating point | `exact/exact.py`, `hierarchy/checks.js` |
| Computation 7.5 | the 18 skeleton walks; parity lemma corroborated; equal to the actual skeletons at levels 2–9 (the computed skeletons record slightly more than Definition 5.3, e.g. rim-run parities) | `hierarchy/induction.js` |
| Proposition 5.4 (check) | reduced walk equals direct composition at all 1,424 parent ports, levels 1–8 | `hierarchy/checks.js` |
| Lemma 8.8 | symbolic label computation, 1,068 + 1,038 obligations; certificates | `hierarchy/certnest.js`, `certify_fib.py`, `certify.py` |
| Lemma 8.8 and Theorem 8.5 | second implementation: symbolic label computation in letter lengths, 150 + 150 distinct obligations, certified; concrete runs for $2 \leq b \leq 40$ follow the same branches; Table 5 reproduced for $2 \leq b \leq 6$; corrupted inputs rejected | `second/nest2.py` |
| Theorem 8.5, Table 5 | strand following in exact arithmetic for $b \leq 3$ to level 7 (used: $b = 1$); in floating point for $b \leq 6$ to level 9, and label computation for $b = 1, 2, 3$ (corroboration) | `exact/exact.py`, `core/hier2.js`, `core/nest.js` |
| corroboration | signature and skeleton periodicity at levels 1–9, all kinds; 0 closed strands at every level | `core/reduce.js`, `hierarchy/synth.js` |
| Computation 7.7 and Lemma 7.8 | every pair of supertiles glued along a side and every triple around a vertex allowed by the matching rule, symbolically for all levels, certified; no closed strand except the Γ-star, which cannot occur; the full compositions; configurations against the matching rule (closed strands found) | `faults/faults.py` |
| Section 3 (drawing) | the drawn ribbons realise the pairing, levels 1–3; shared edges glue $e_i$ to $e_j$ with $i + j$ odd, levels 1–4 | `drawing/checkmodel.js`, `exact/exact.py` |
| Remark 3.6 | the drawn pieces: containment, the single crossing, angles at the edges | `drawing/drawcheck.py` |

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